

Anomaly Detection

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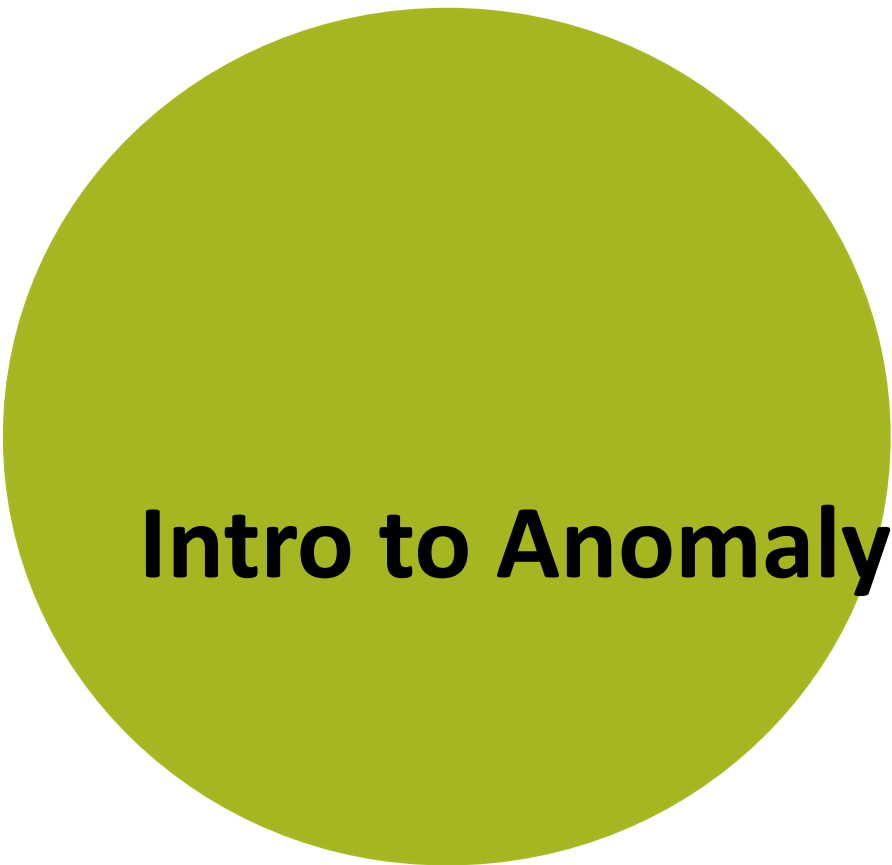
Associate professor, Lab Head



Lecture plan

- Intro to Anomaly detection
- Unsupervised approaches for Anomaly detection. General idea
- Autoencoders for Anomaly detection
- GAN-based Anomaly detection
- Anomaly detection for Time Series





Intro to Anomaly Detection

Problem statement

The problem is to find objects that *anomalous* given training data.

It is ill-specified: small number of labels / no labels / ambiguous labels.

Normal data

Image Type	No Cat	Cat not on approach	Cat on approach	Cat with prey
Count of Images	6,542	9,504	6,689	260
Example				

Anomalous data
1


<https://www.theverge.com/tldr/2019/6/30/19102430/amazon-engineer-ai-powered-catflap-prey-ben-hamm>

Problem examples are numerous

- Fraud detection 🕵️
- Failure detection for an airplane ✈️
- Intrusion detection 😱
- Earthquake prediction 💣



Problem examples

- Fraud detection 🕵️
- Failure detection for an airplane ✈️
- Intrusion detection 😱
- Earthquake prediction 💣



Typical challenges:

- Requires problem-specific knowledge => new problem – new approach
- Hard to identify something we don't see
- Bunch of various problem statements => how to define what is anomaly?



Anomaly type taxonomy

Normal data

Image Type	No Cat	Cat not on approach	Cat on approach	Cat with prey
Count of Images	6,542	9,504	6,689	260
Example				

Point Anomaly



Group Anomaly



Contextual Anomaly



In 2019



He is mad! We should avoid him. (anomaly)

In 2020



He takes care of himself and others. Well done! (normal)

Different taxonomies

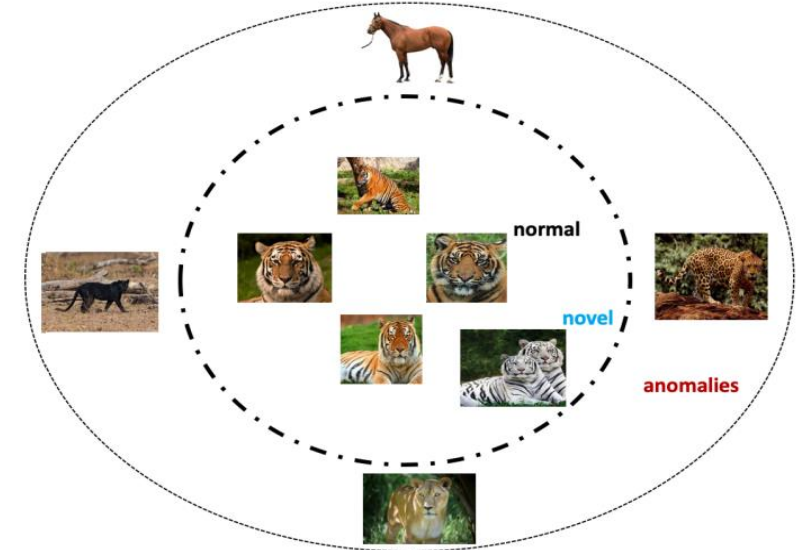
By method:

- Supervised
- Unsupervised
- Semi-supervised

By problem statement:

- Anomaly detection
- Novelty detection

Anomaly Detection



By data:

- Structured data
- Sequential data
- Unstructured data
- Semi-structured data

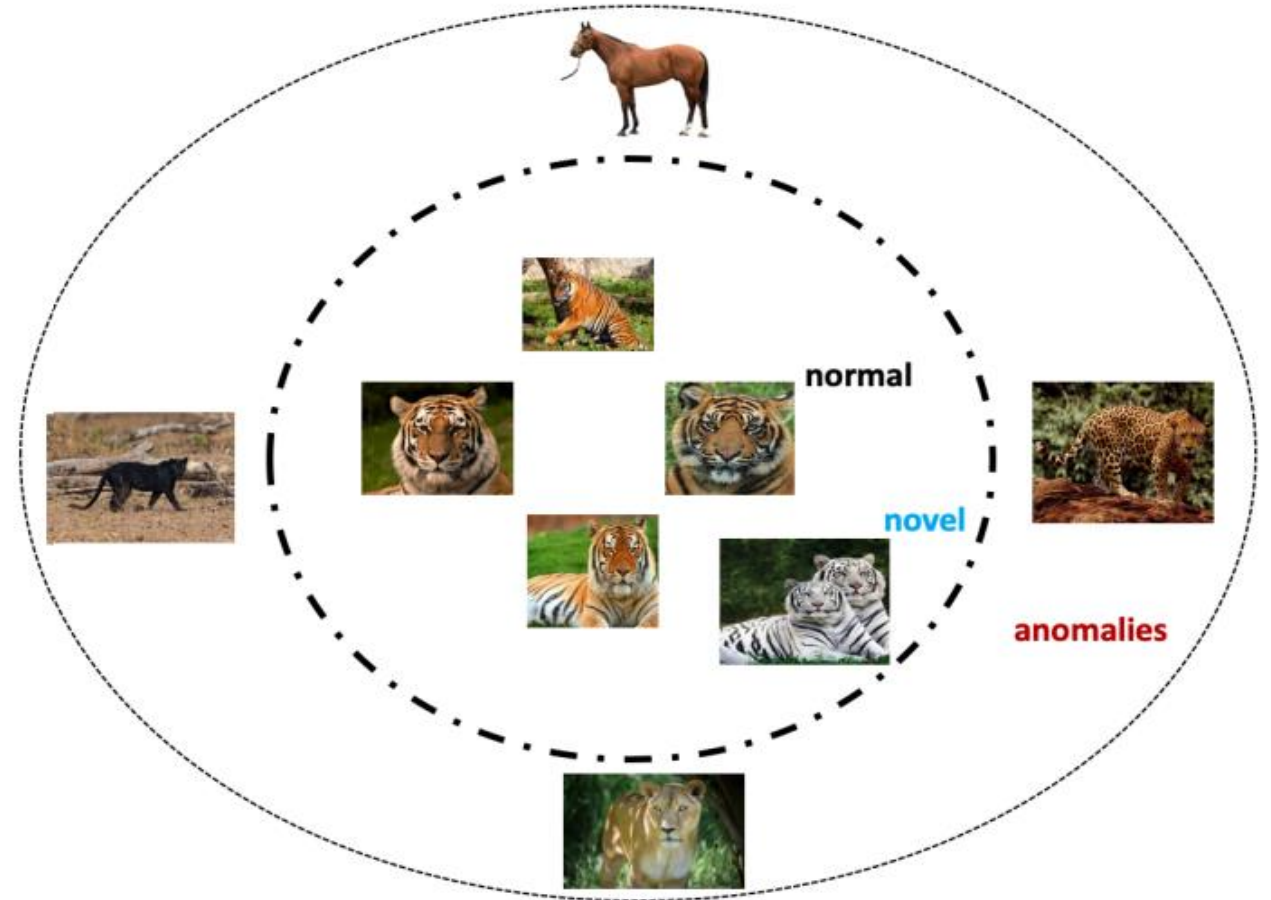
Taxonomy by type of anomalies

Novelty detection

We have never seen the objects of this class. They are **not** in the training sample. One-class setting.

Anomaly detection

Outliers, can be already in the training sample: labeled or unlabeled



Well-defined anomaly assumption – a more formal approach

WDAD [Tom Dietterich]: the anomalies are drawn from a well-defined probability distribution

Example: repeated instances of known machine failures

The WDAD assumption is often risky:

- adversarial situations: fraud, insider threats, cyber security
- diverse set of potential causes, novel device failure modes
- user's notion of “anomaly” changes with time.

Example: anomaly is an “interesting point”



https://images.prismic.io/sketchplanations/54100eae-1fee-40b7-b3d7-c0b9e309da1c_SP+711+-+Looking+under+the+lamppost.png?auto=compress,format



Supervised anomaly detection

Supervised Anomaly detection is an imbalanced classification problem

Main idea: correct the balance of classes, as most ML methods assume uniform prior over classes

Weights for classes

- *Not helpful in most cases*

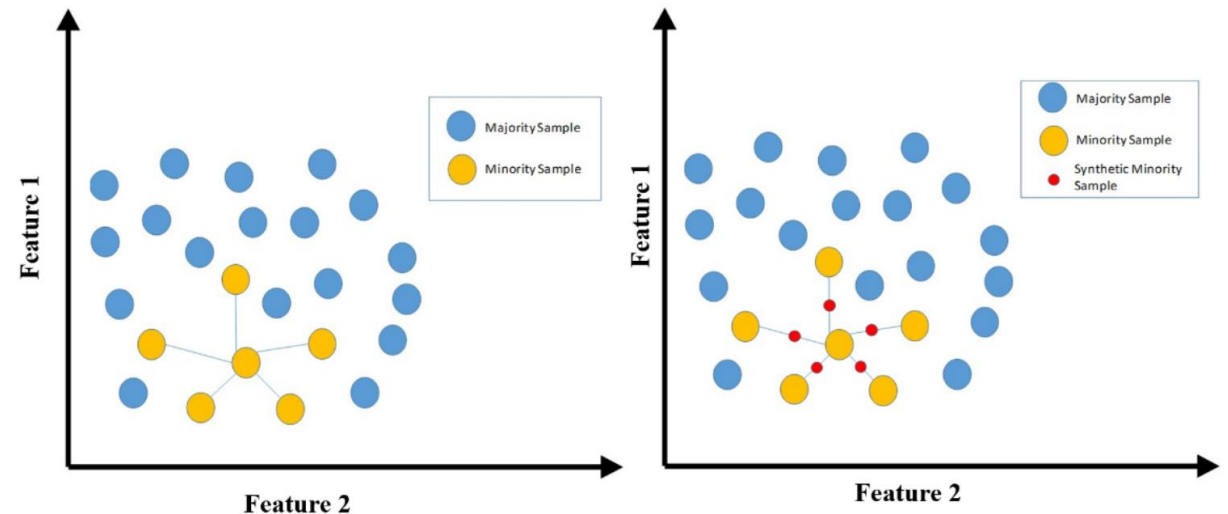
Resampling methods

- *Undersampling*
- *Oversampling/data generation: SMOTE, etc.*
- *Generative models*

Specific models

SMOTEboost, DeepSMOTE

Specific metrics: F1, PR AUC



SMOTE resampling

Practical questions

How to choose which method to use?

How to choose a resampling parameter?

Cross-validation (but it is harder here,
imagine only 10 anomalies available)

So: an ad-hoc validation scheme

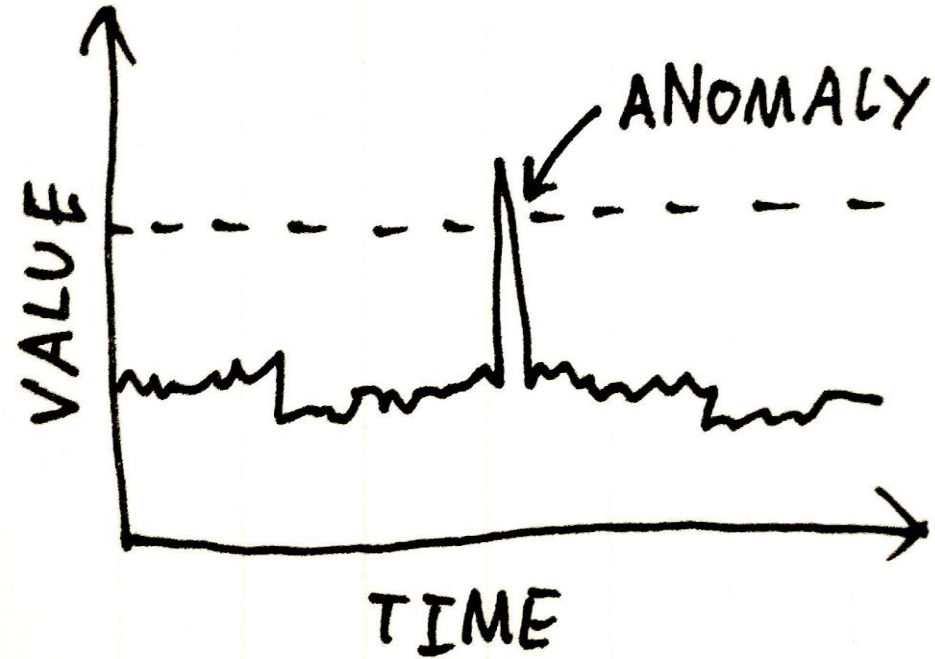


Unsupervised density-based Anomaly Detection

Classic pipeline for anomaly detection

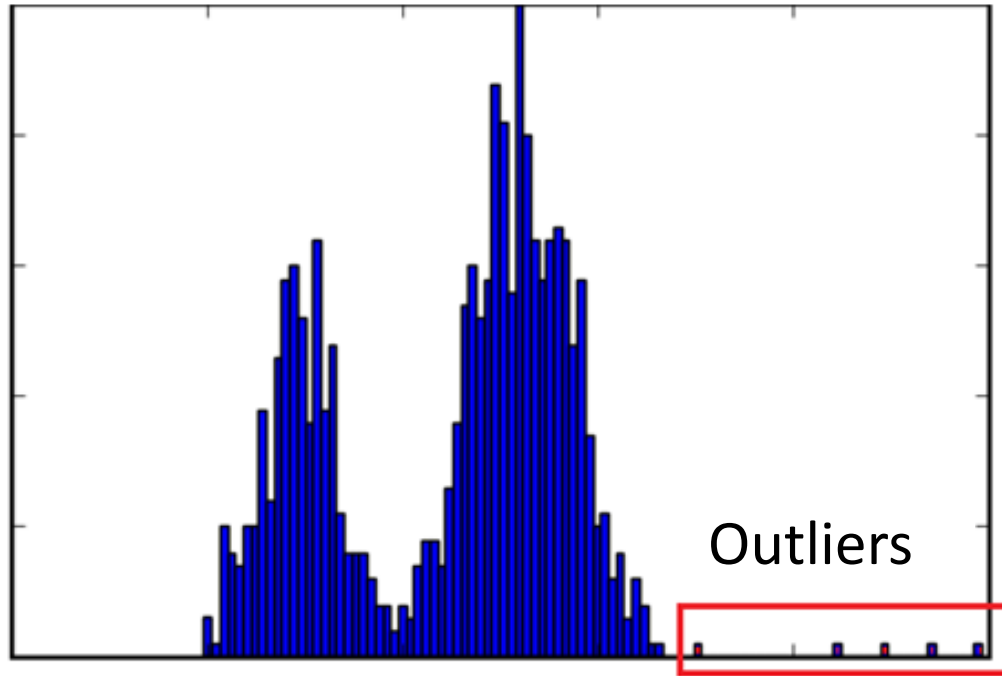
1. Construct an anomaly score $s(x)$ using data x
2. Signal about anomaly if the anomaly score is greater than a threshold τ

The threshold selection τ is a separate problem, as we often have only positive examples

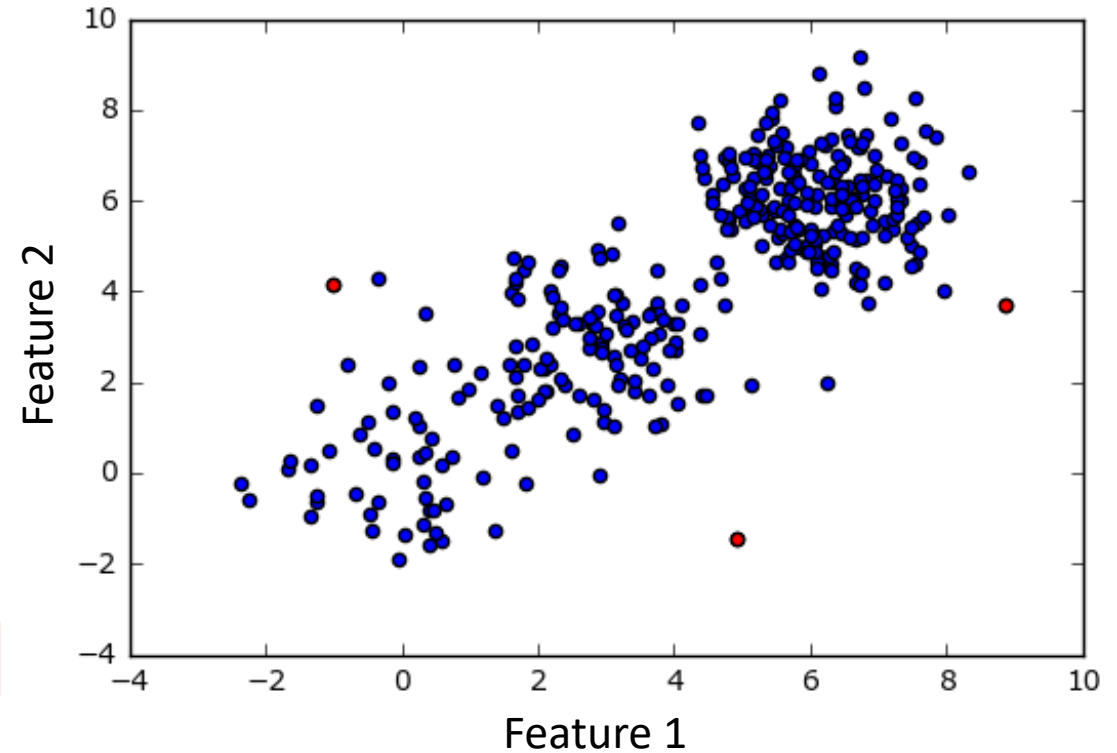


Anomaly detection with a keen eye

One feature anomaly



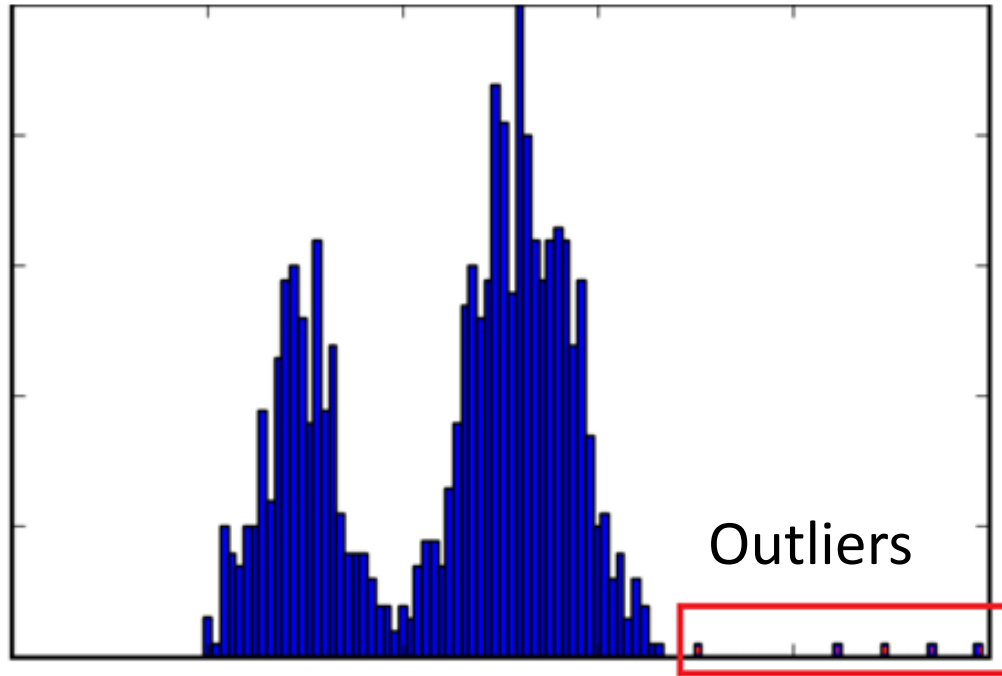
Pair of features anomaly



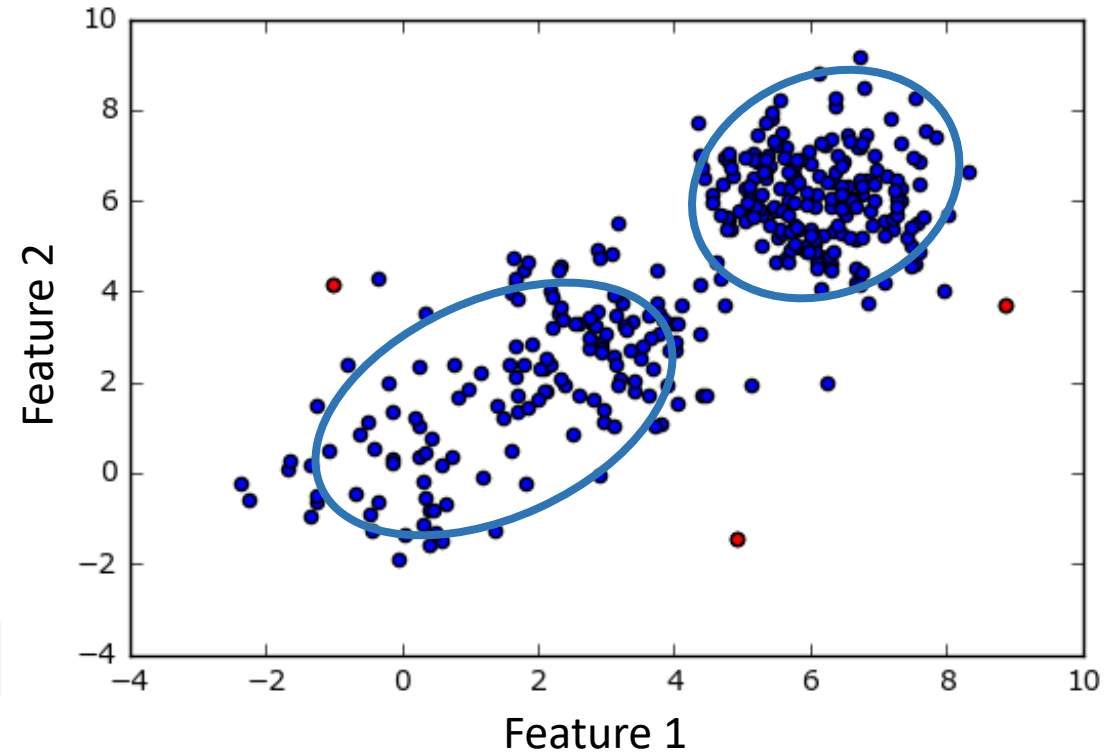
- – Anomaly
- – Normal data

Anomaly detection with a keen eye

One feature anomaly



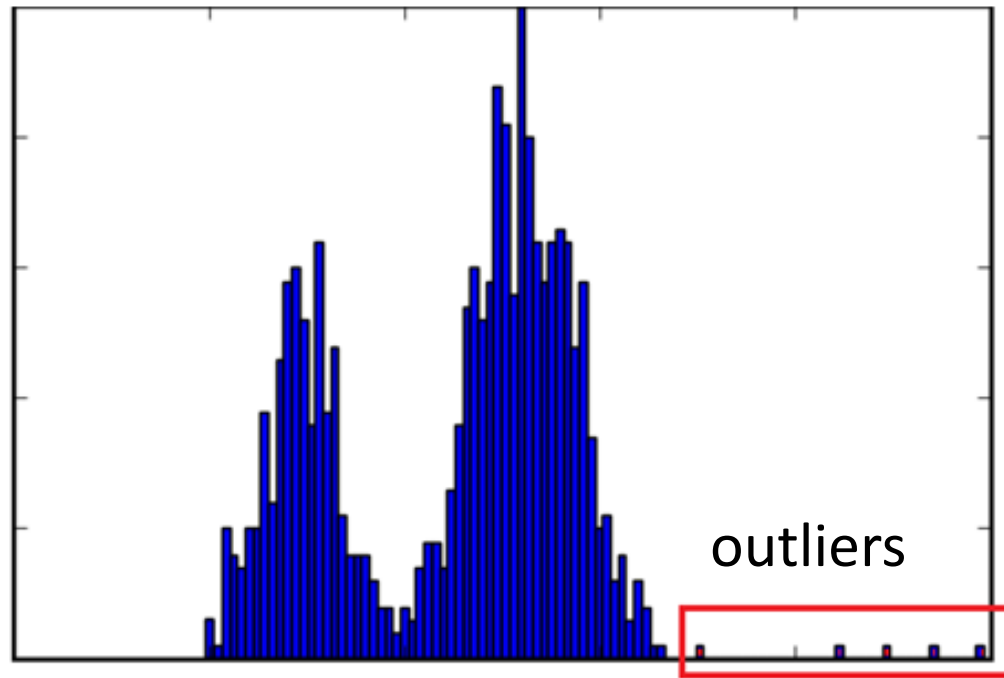
Pair of features anomaly



- – Anomaly
- – Normal data

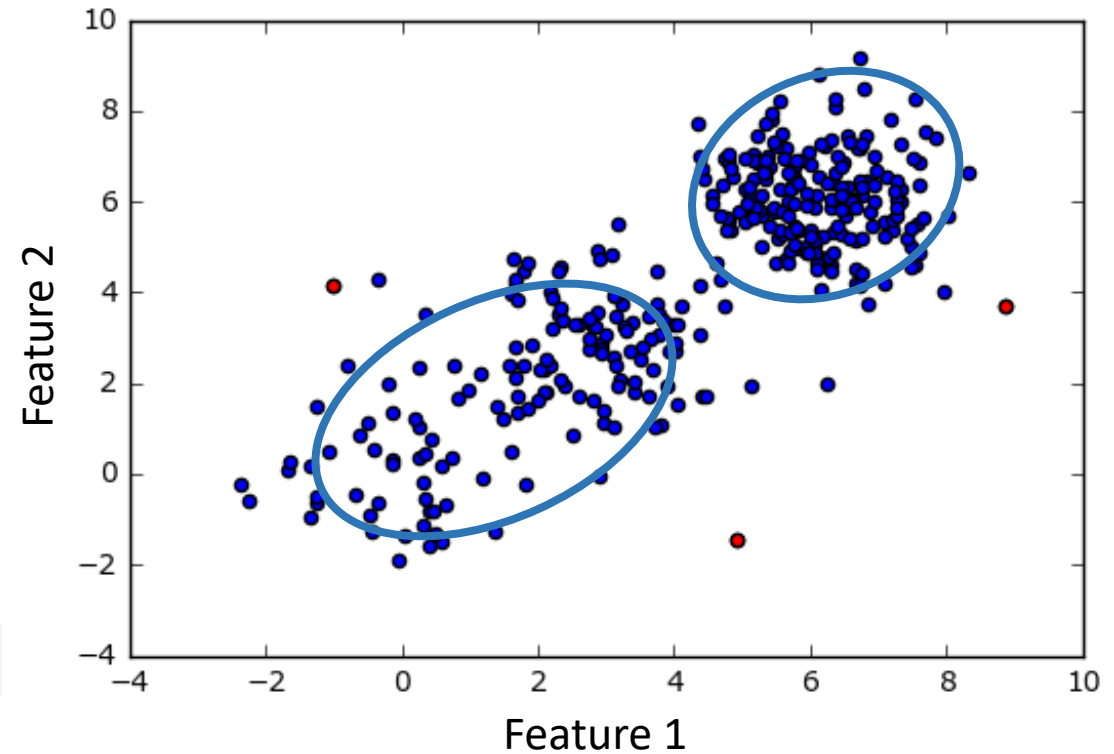
Formal interpretation: low density corresponds to outliers

One feature anomaly



$\hat{p}(x)$ is small

Pair of features anomaly



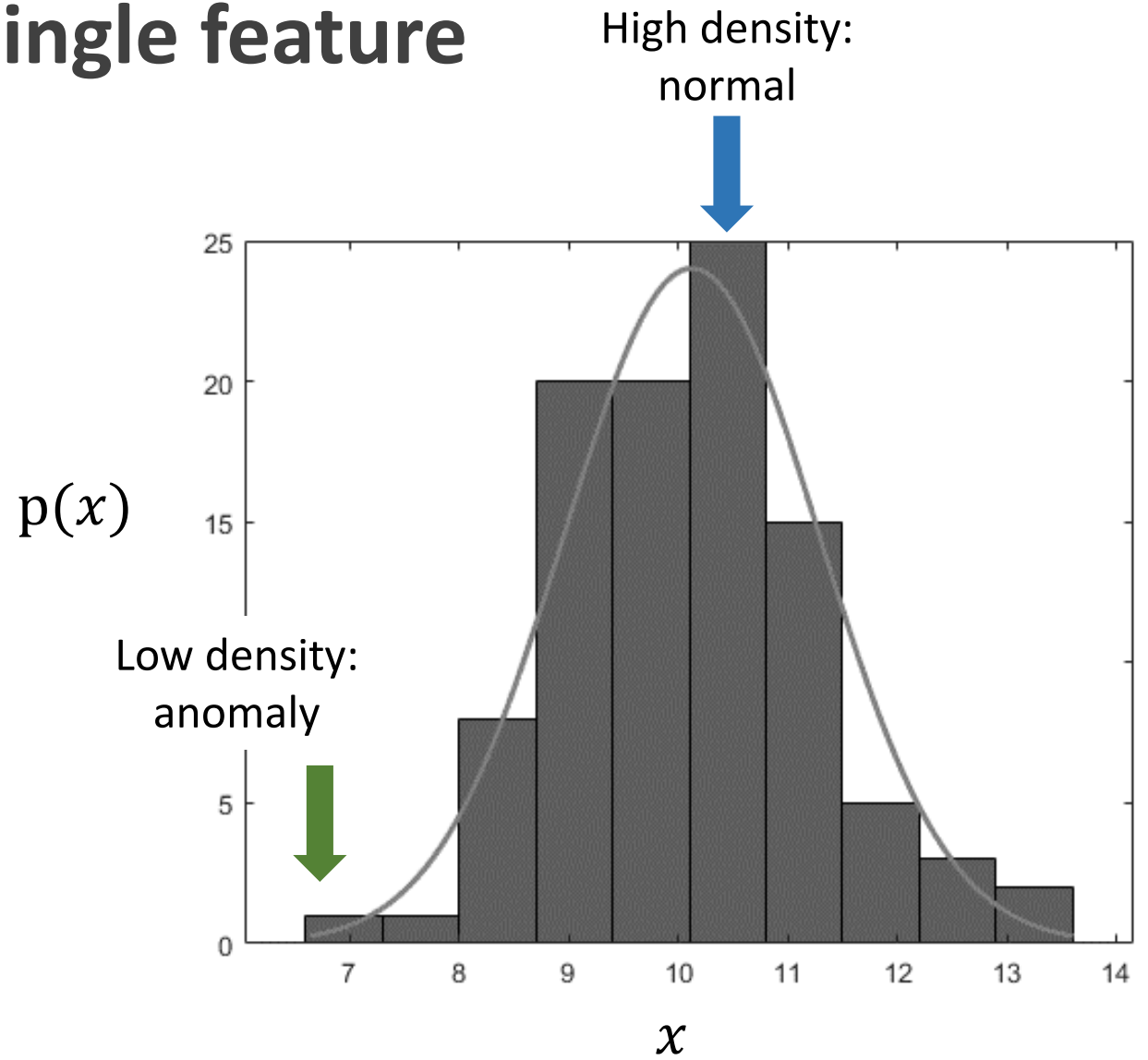
- – anomaly
- – normal data

Anomaly detection for a single feature

Histogram: a step-wise density estimation

Parametric approach: the curve is the density estimation $\hat{p}(x)$ for data from a parametric family

Example: Gaussian density
 $N(\mu, \sigma^2)$



Kernel density estimation

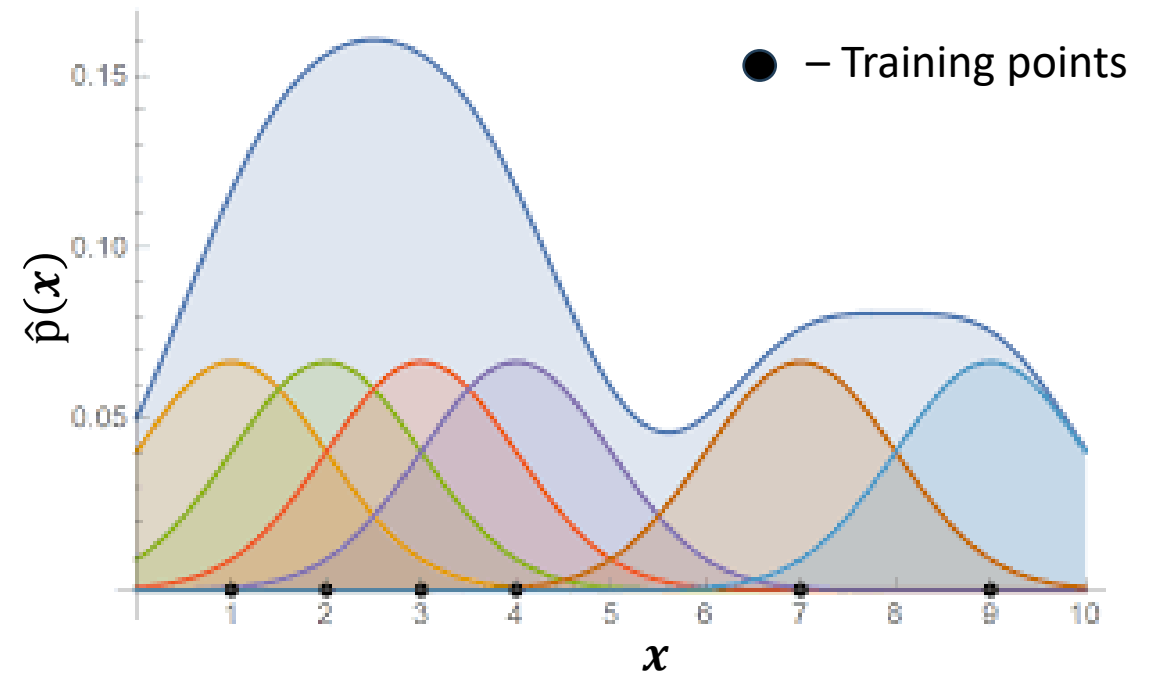
Estimate the density $p(\mathbf{x})$ for a low input dimension easily

A kernel method is a key:

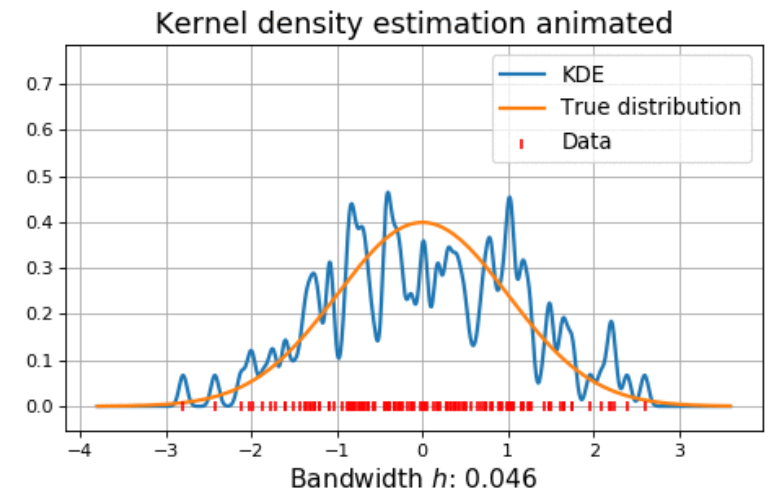
$$\hat{p}(\mathbf{x}) = \frac{1}{n} \sum_i k(\mathbf{x} - \mathbf{x}_i),$$

$$k(\mathbf{x} - \mathbf{x}') = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{|\mathbf{x} - \mathbf{x}'|^2}{2\sigma^2}\right)$$

We can vary kernel width σ to maximize the fit quality. *Out of scope of today's lecture, theoretical results exist*



<https://ekamperi.github.io/math/2020/12/08/kernel-density-estimation.html>



https://github.com/fbeilstein/machine_learning/blob/master/lecture_15_kernel_density_estimation.ipynb

Anomaly detection using a pair of features

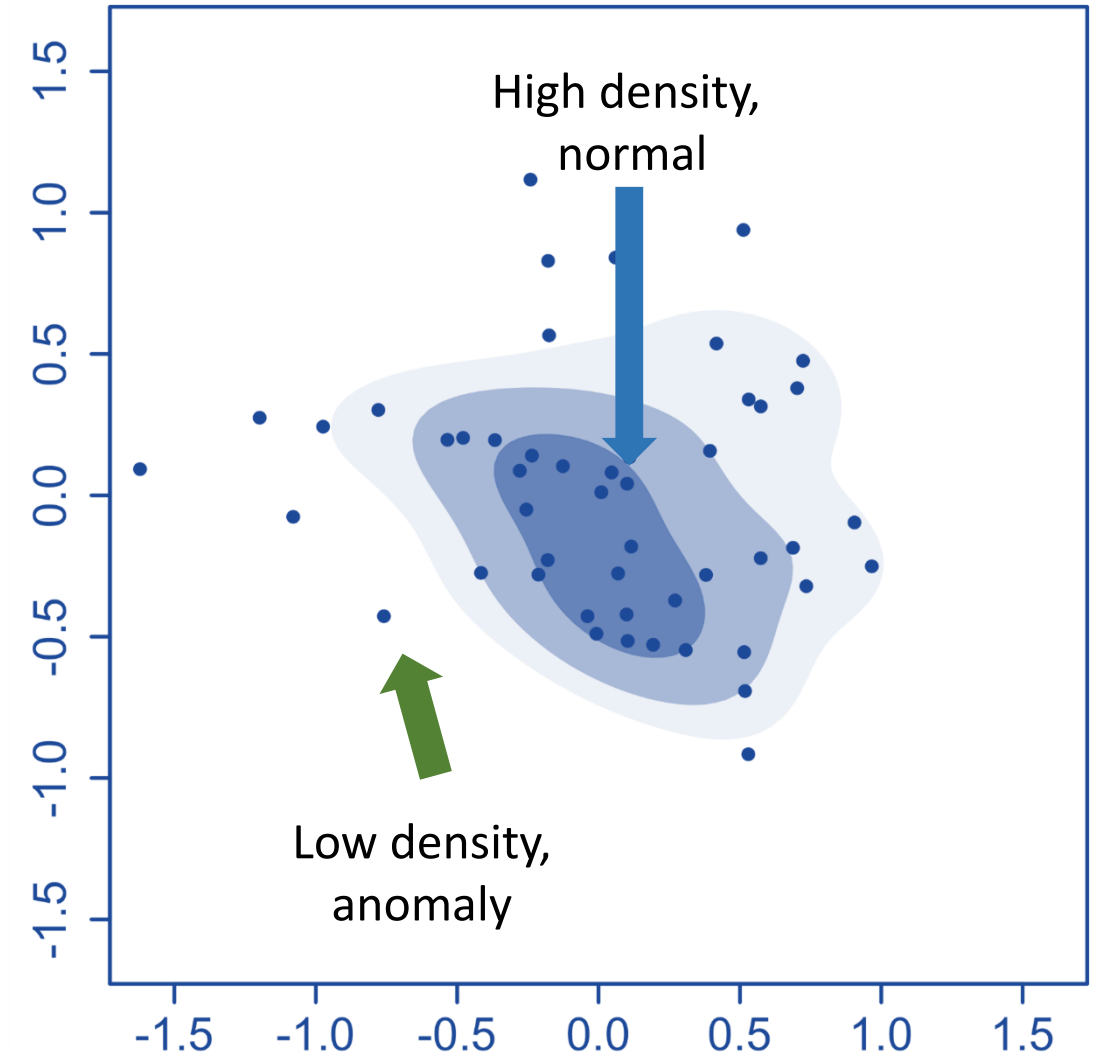
Points are real data

The color intensity corresponds to an estimated density value

$$\hat{p}(\mathbf{x}) = \frac{1}{n} \sum_i k(\mathbf{x} - \mathbf{x}_i),$$

$$k(\mathbf{x} - \mathbf{x}') = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{|\mathbf{x} - \mathbf{x}'|^2}{2\sigma^2}\right)$$

Can we perform KDE in a higher-dimensionality input space?



Theoretical results for high-dimensional KDE

For a twice-differentiable density

$$MISE \approx n^{-\frac{4}{4+d}}$$

Exponential increase:

$$d = 1: n^{-4/5}$$

$$d = 4: n^{-1/2}$$

$$d = 12: n^{-1/4}$$

MISE – the mean integrated squared error for density

n is the sample size

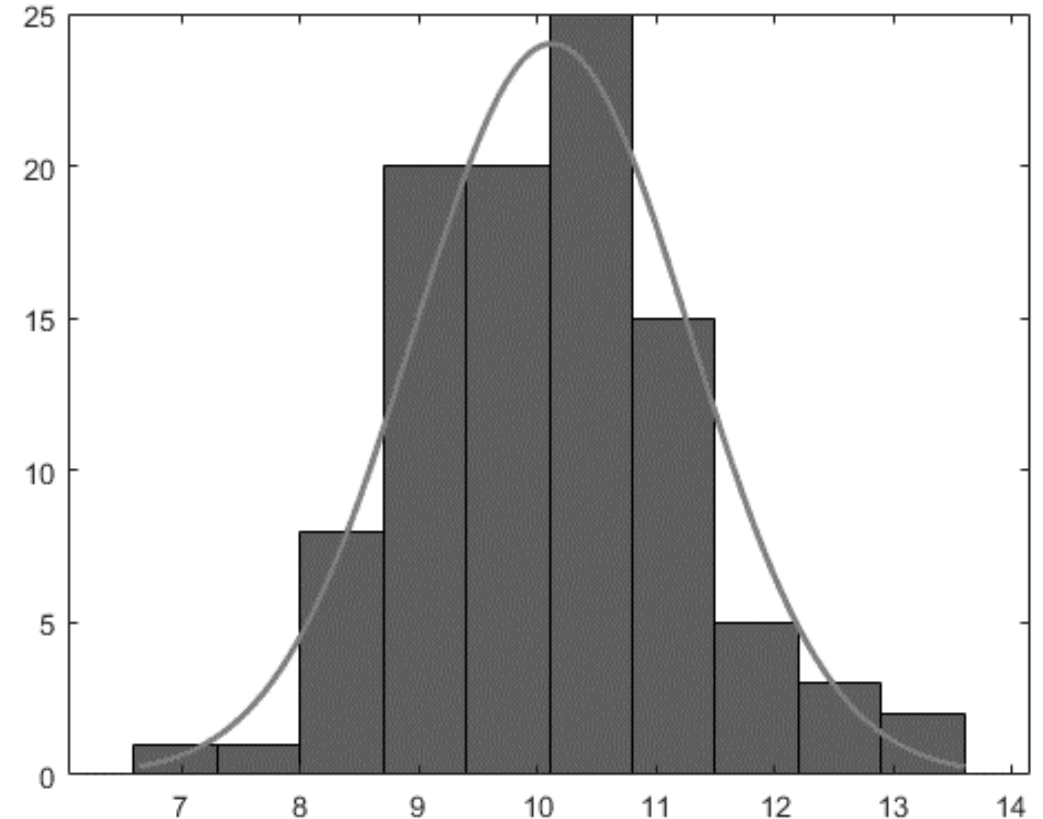
d is the data dimensionality

LODA anomaly detector

1. Build M random linear projections
2. Estimate density for each projection $\hat{p}_i(\mathbf{x}), i = 1, \dots, M$
3. Mean density for all projections

$$S(\mathbf{x}) = -\frac{1}{M} \sum_{i=1}^M \log \hat{p}_i(\mathbf{x})$$

Get mean «surprise» $\mathbb{E} S(\mathbf{x})$

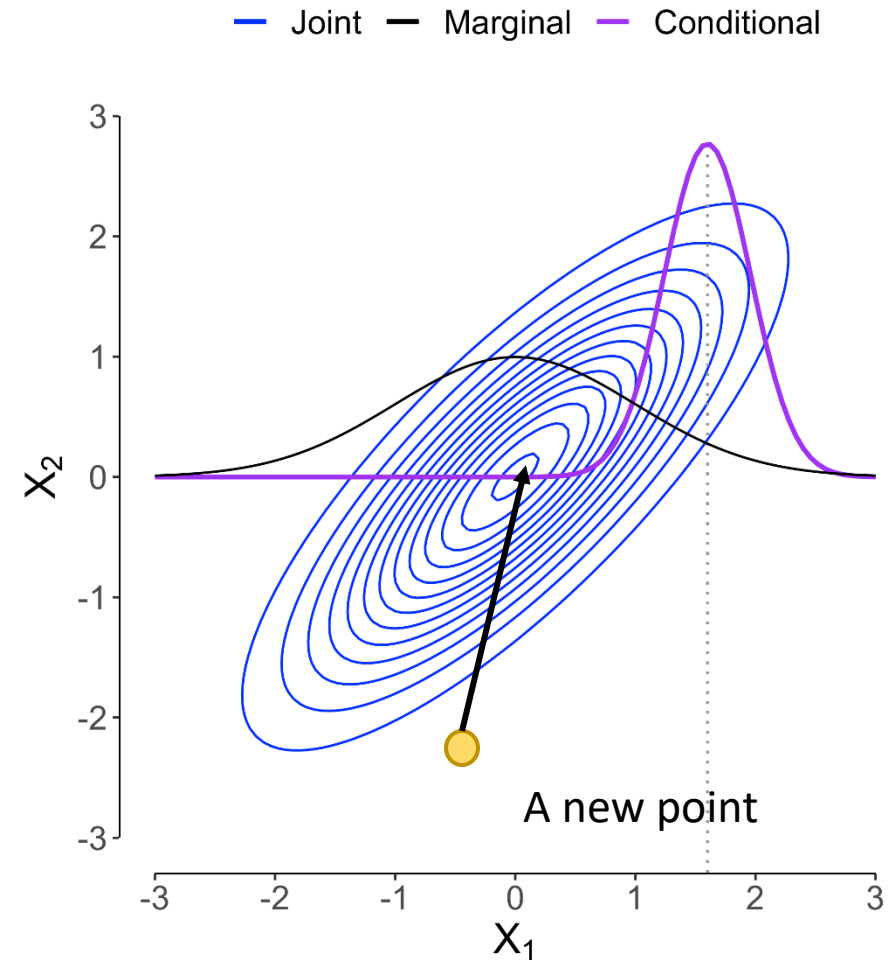


Parametric Gaussian estimator

Parametric approach: the curve is the density estimation $\hat{p}(\mathbf{x})$ for data from a parametric family

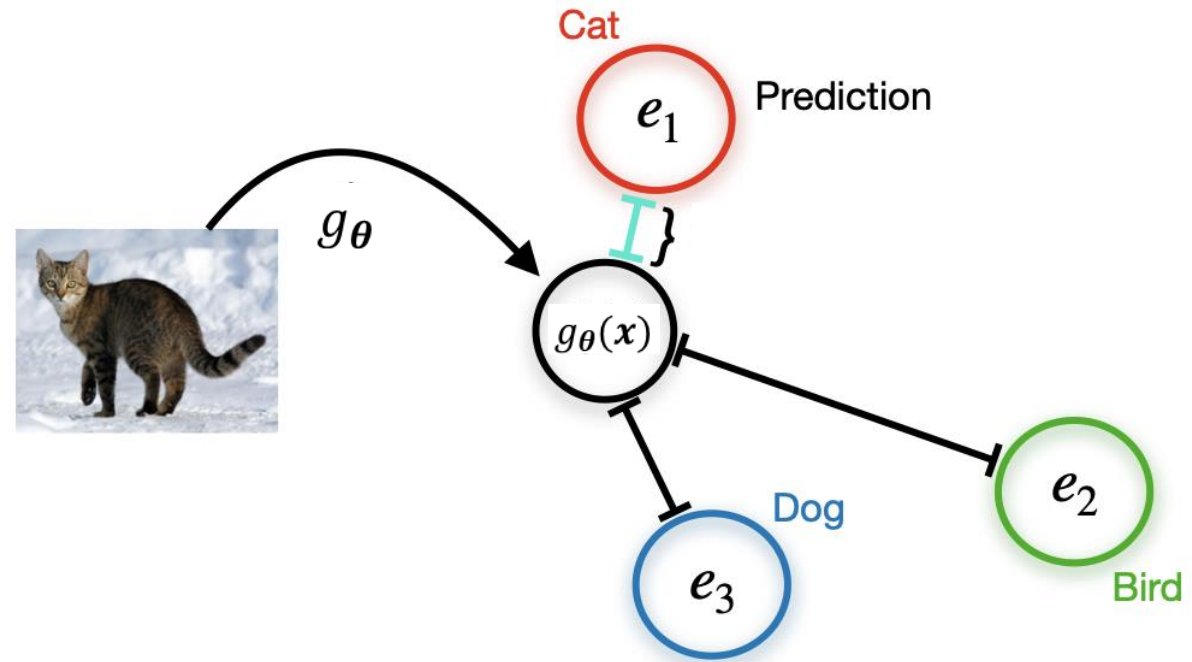
Example: Gaussian density $N(\mu, \sigma^2)$

Estimate μ and σ^2 from data: a variant of the maximum likelihood approach

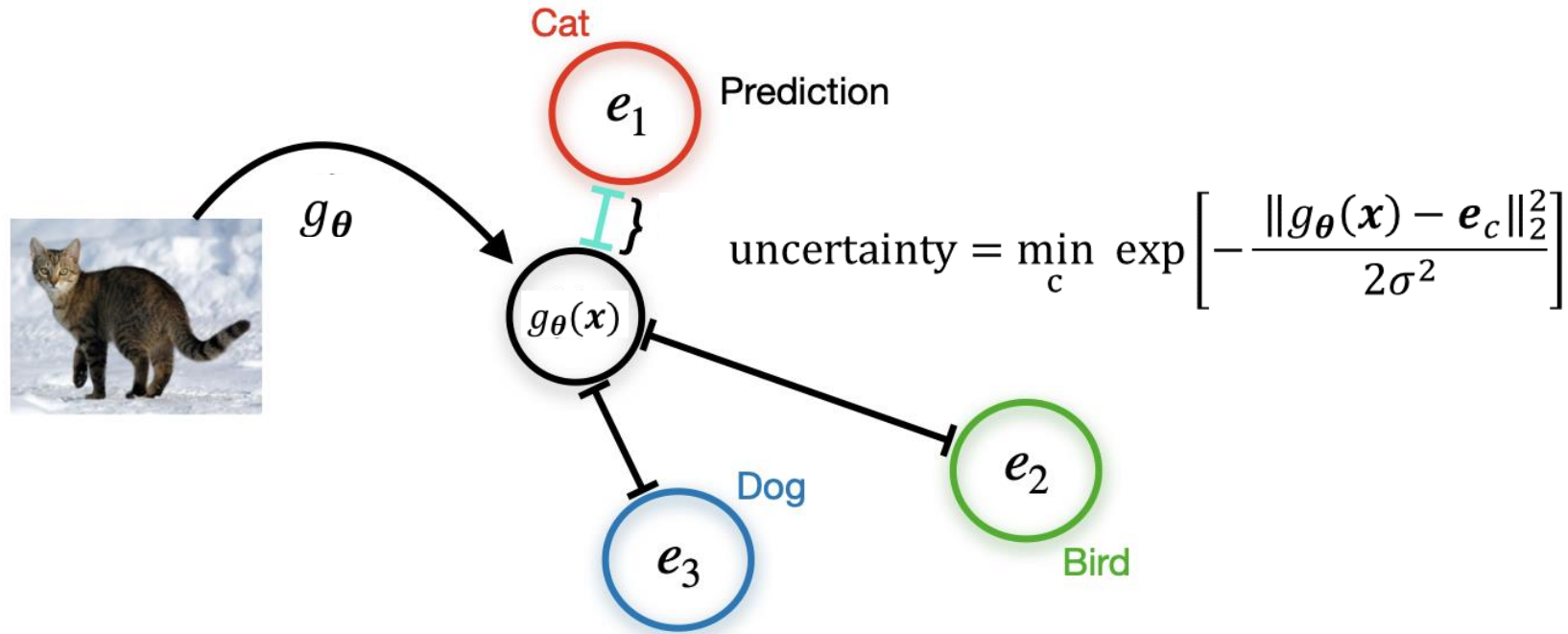


Representations from Neural Networks

- $g_{\theta}(x)$ is a neural network backbone
- e_i is the mean embedding for i-th class



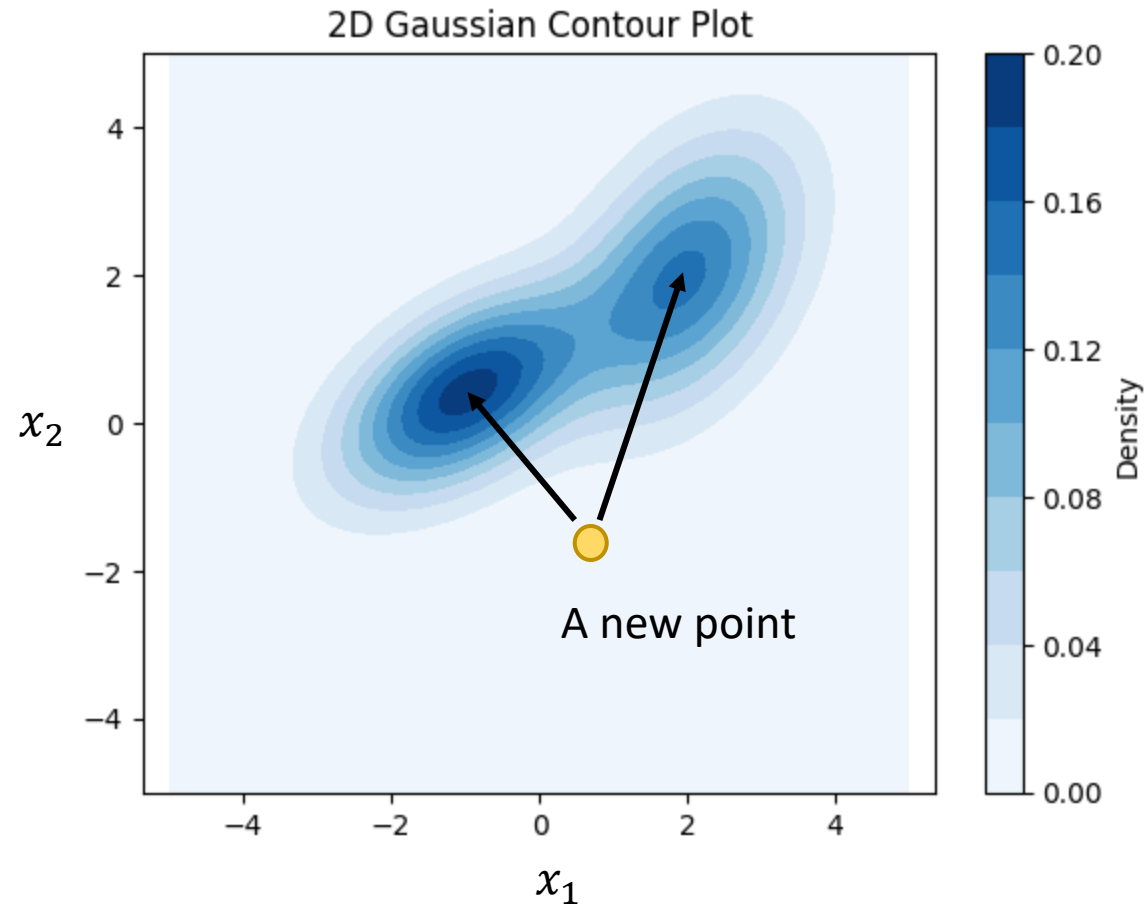
Utilize representations for anomaly detection



A DUQ architecture: (1) the input is mapped to the feature space, where (2) it is assigned to the closest centroid. (3) The distance to this centroid is uncertainty/anomaly score.

Source: Van Amersfoort, Joost, et al. "Uncertainty estimation using a single deep deterministic neural network." *ICML*, 2020.

Probabilistic interpretation: Gaussian mixture density

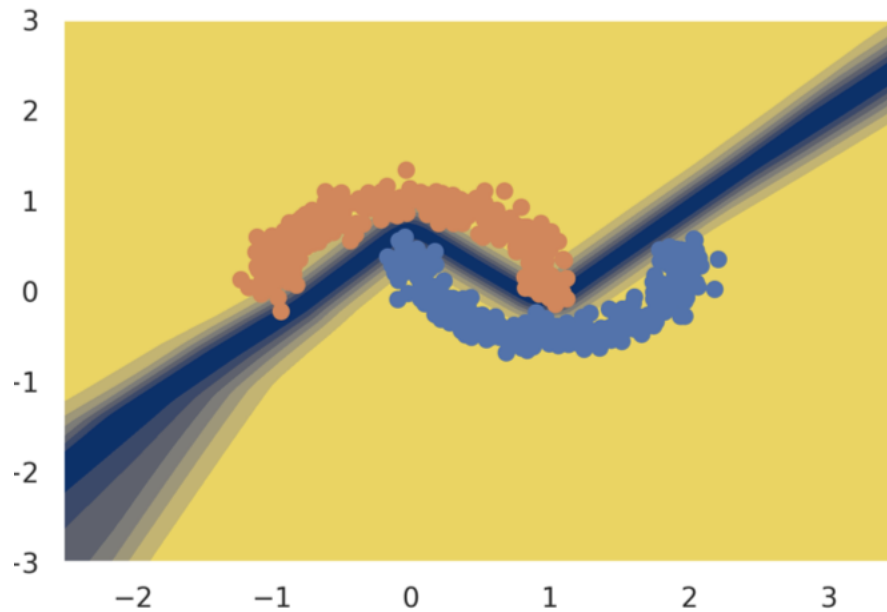
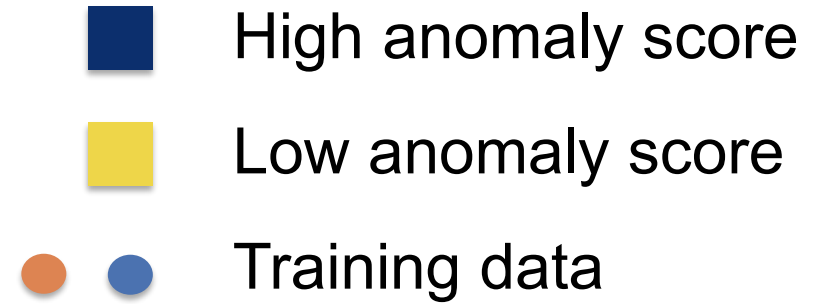


$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

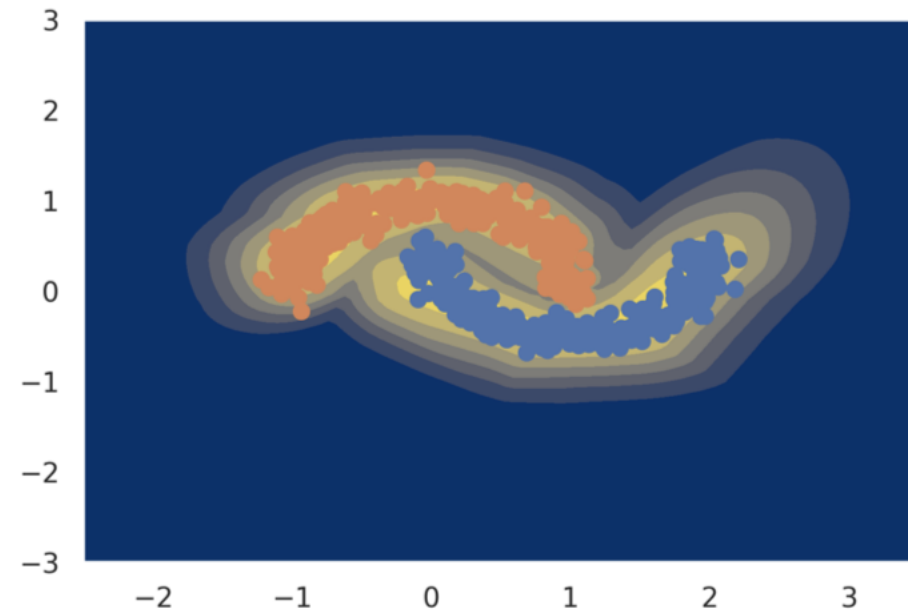
$$\mathcal{N}(x \mid \mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2} |\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_k)^\top \Sigma_k^{-1} (x - \mu_k)\right)$$

For the high input dimension d ,
only the closest center matters

Toy example for DUQ



Baseline



DUQ model



Forest-based Anomaly Detection

Top recommendations from the three studies

- Isolation based methods: iForest and iNNE [8]
 - All known weaknesses of iForest are overcome by iNNE, with higher time cost; but still has significantly lower time cost than kNN.
 - Nearest neighbour-based methods: aNNE and kNN
 - Clustered anomalies: ABOD & LOF
 - iNNE [5] can do well in detecting clustered anomalies.
 - Kernel Mahalanobis [6]: The key weakness is the time cost
-
- Among the compared methods, iForest and iNNE have the highest detection accuracy and also have the lowest time cost.

[6] Hoffmann, H. (2007). Kernel PCA for Novelty Detection, Pattern Recognition, 40(3), 863–874.

[8] Bandaragoda, T. R., Ting, K. M., Albrecht, D., Liu F. T., Wells, J. R. (2018). Isolation-based Anomaly Detection using Nearest Neighbour Ensembles. Computational Intelligence. Doi:10.1111/coin.12156.

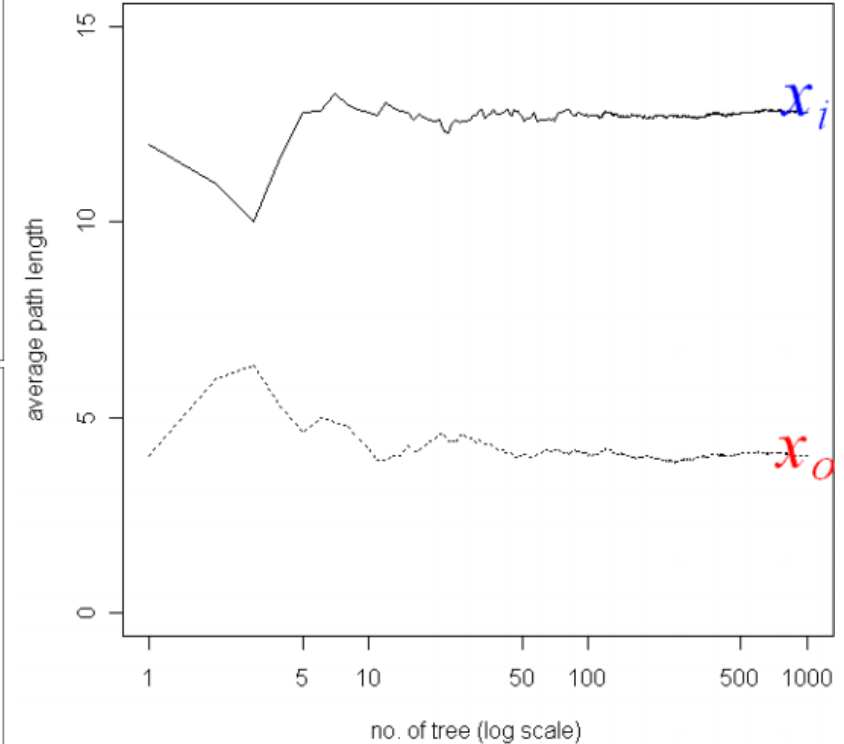
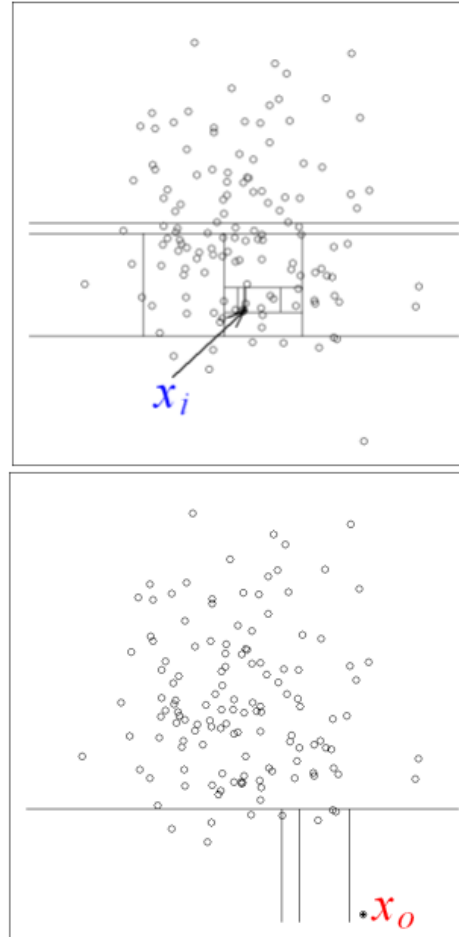
Models based on ensembles of decision trees

- Isolation forest iForest
- Isolation Nearest Neighbours Ensembles INNE

https://federation.edu.au/_data/assets/pdf_file/0011/443666/ICDM2018-Tutorial-Final.pdf

Isolation tree

1. Repeat for k trees: draw random splits until a node is separated
2. Calculate an average path length
3. It is the normality score



Source: Liu et al 2008

[Liu et al ICDM 2008]

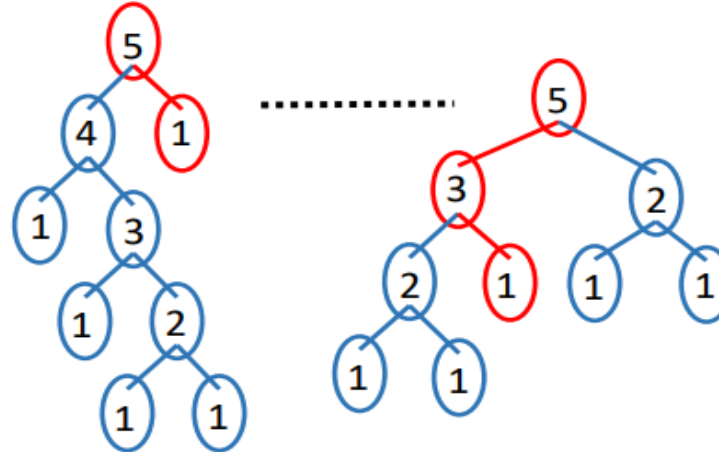
https://federation.edu.au/__data/assets/pdf_file/0011/443666/ICDM2018-Tutorial-Final.pdf

Isolation forest

- A collection of isolation trees (iTrees)
- Each iTree isolates every instance from the rest of the instances in a given sample
- Anomalies are ‘few and different’
 - More susceptible to isolation
 - Shorter average path

$$Score(\mathbf{x}) = \frac{1}{t} \sum_{i=1}^t \ell_i(\mathbf{x})$$

where $\ell_i(\mathbf{x})$ is the path length of $\mathbf{x}$ traversed in tree i



https://federation.edu.au/__data/assets/pdf_file/0011/443666/ICDM2018-Tutorial-Final.pdf

[Liu et al ICDM 2008]

INNE approach

Let $D \subset \mathbb{R}^d$ be a given data set, and let $\|a - b\|$ denote the Euclidean distance between a and b , where $a, b \in \mathbb{R}^d$.

Let $\mathcal{S} \subset D$ be a subsample of size ψ selected randomly without replacement from a dataset $D \subset \mathbb{R}^d$; and η_x be the nearest neighbour of x .

Definition 1: A hypersphere $B(c)$ centred at c with radius $\tau(c) = \|c - \eta_c\|$, is defined to be $\{x : \|x - c\| < \tau(c)\}$, where $x \in \mathbb{R}^d$ and $c, \eta_c \in \mathcal{S}$.

Definition 2: Isolation score for $x \in \mathbb{R}^d$ based on $\mathcal{S}$ is defined as follows:

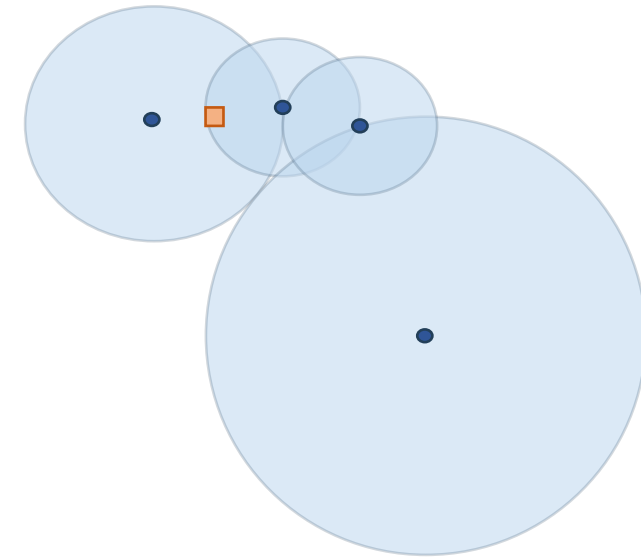
$$I(x) = \begin{cases} 1 - \frac{\tau(\eta_{cnn(x)})}{\tau(cnn(x))}, & \text{if } x \in \bigcup_{c \in \mathcal{S}} B(c) \\ 1, & \text{otherwise} \end{cases}$$

where $cnn(x) = \arg \min_{c \in \mathcal{S}} \{\tau(c) : x \in B(c)\}$.

Definition 3: iNNE has a set of t sets of hyperspheres, generated from t subsamples $\mathcal{S}_i$, defined as follows:

$$\left\{ \left\{ B(c) : c \in \mathcal{S}_i \right\} : i = 1, \dots, t \right\}$$

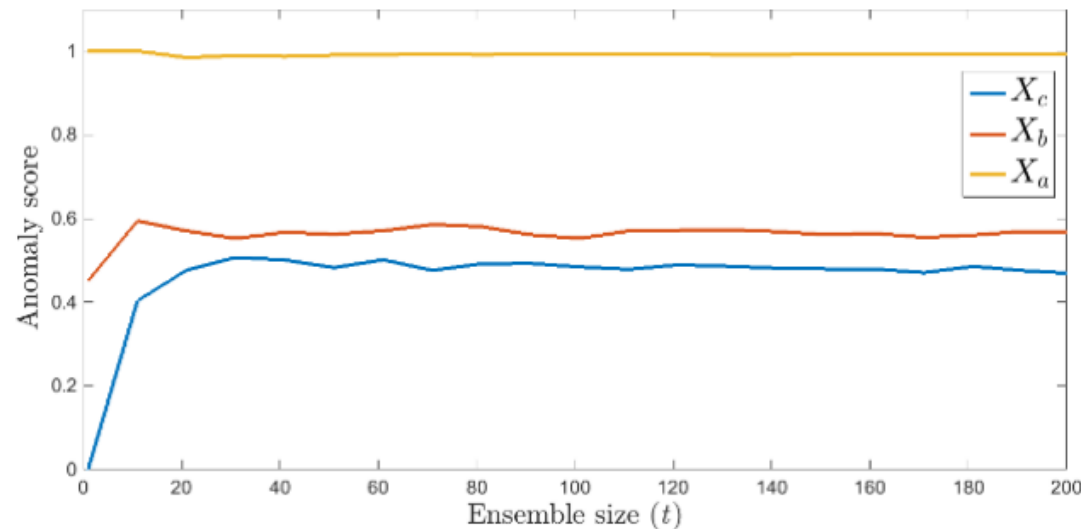
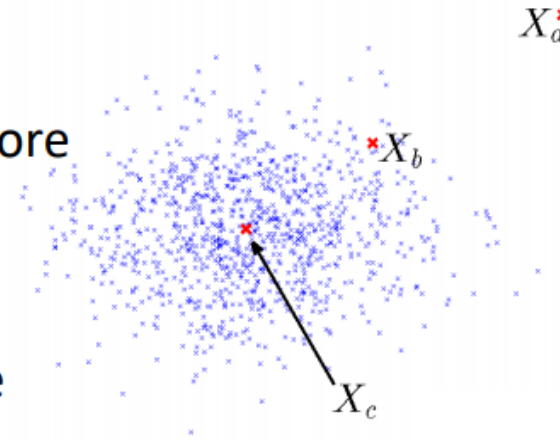
The final anomaly score is the mean value of subsample anomaly scores

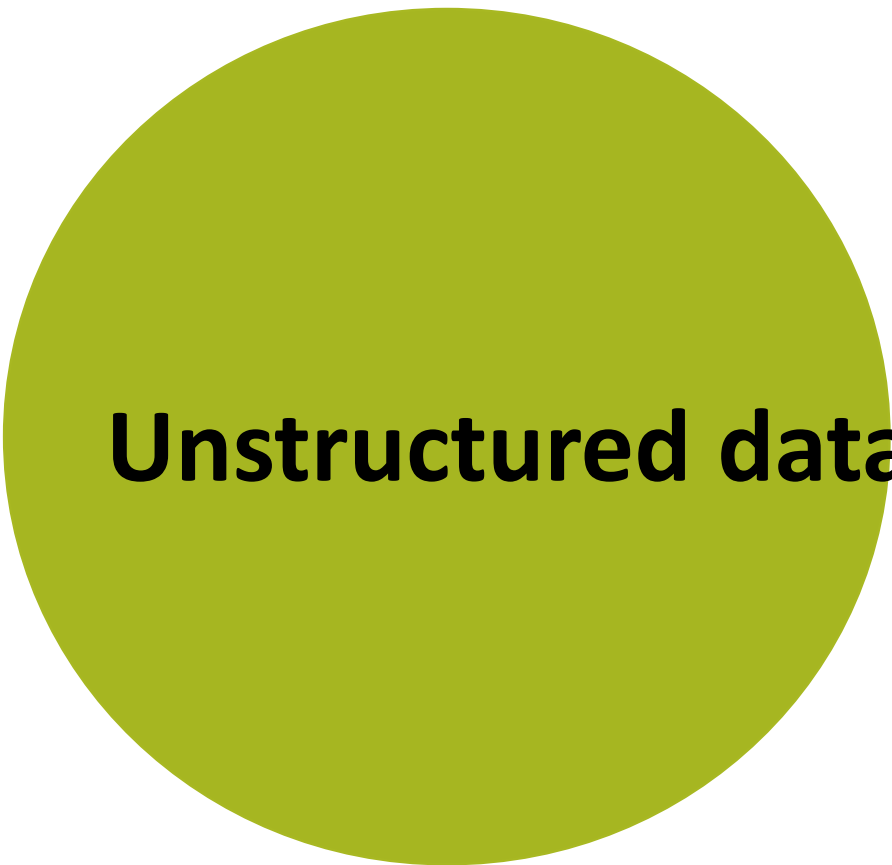


https://federation.edu.au/__data/assets/pdf_file/0011/443666/ICDM2018-Tutorial-Final.pdf

Numerical example

- X_a has the maximum anomaly score
- X_b has a lower anomaly score
- X_c has the lowest anomaly score





Unstructured data anomaly detection

Classic approach revisited

- A sample $D = \{\mathbf{x}_i\}_{i=1}^n$ is given, each $\mathbf{x} \in \mathbb{R}^d$.
- Construct models

$$\hat{x}_1 = f_1(x_2, x_3, \dots, x_d),$$

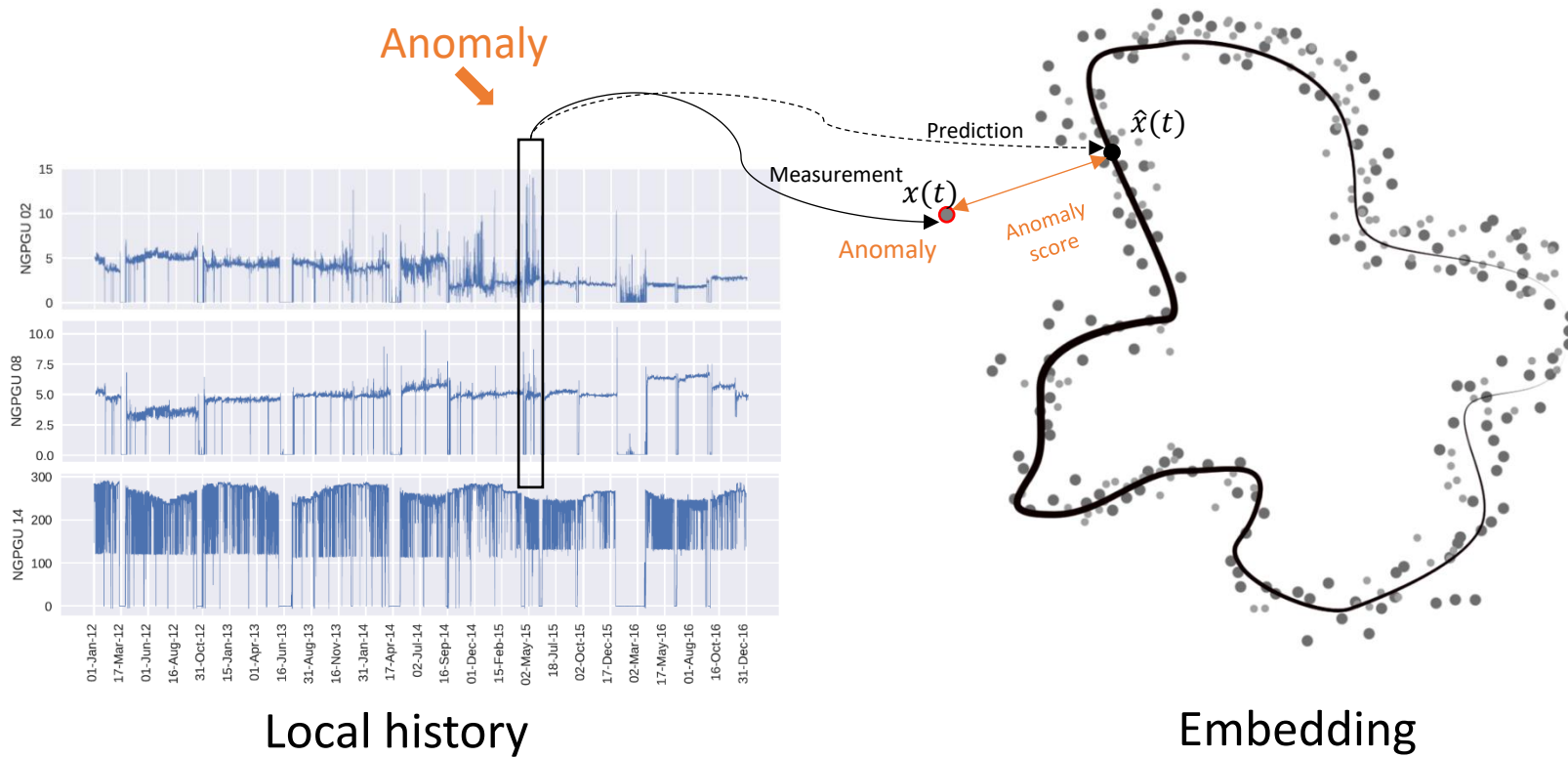
...

$$\hat{x}_d = f_d(x_1, x_2, \dots, x_{d-1}).$$

- We have d anomaly scores for $\mathbf{x} = \{x_1, x_2, \dots, x_d\}$:

$$s_i(\mathbf{x}) = |\hat{x}_i - x_i|, i = \overline{1, d}.$$

Autoencoder anomaly detection. General approach



Unsupervised anomaly detection. General approach

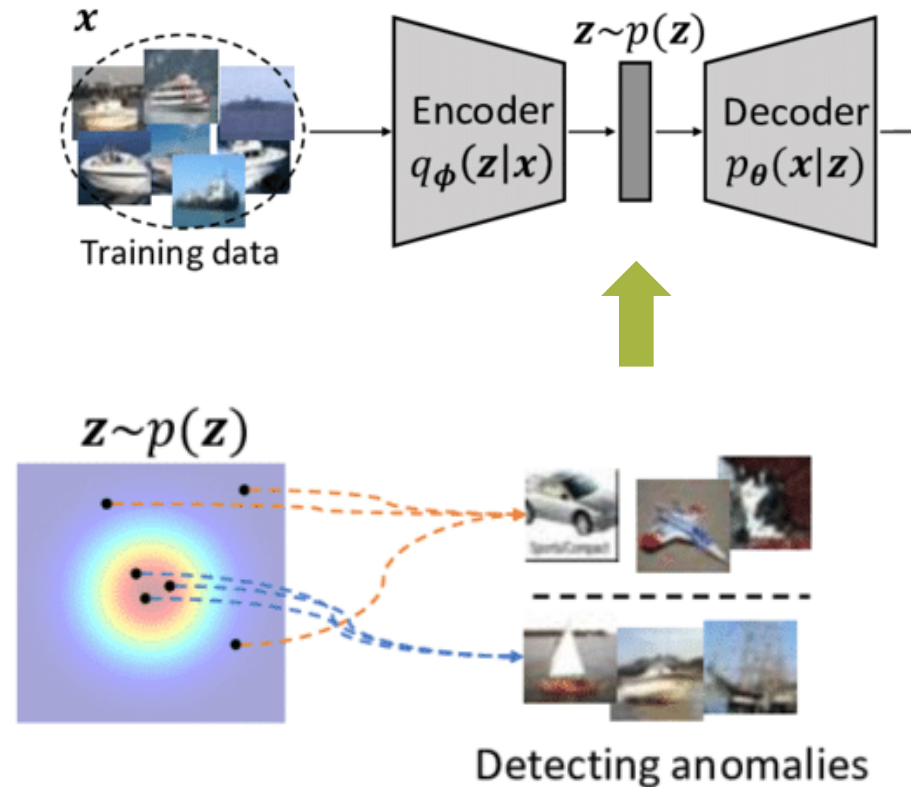
- A sample $D = \{\mathbf{x}_i\}_{i=1}^n$ is given, each $\mathbf{x} \in \mathbb{R}^d$.
- Construct encoder and decoder model

$$\mathbf{z}_i = e(\mathbf{x}_i),$$

$$\mathbf{x}_i \approx \hat{\mathbf{x}}_i = d(\mathbf{z}_i) = d(e(\mathbf{x}_i)).$$

- We have an anomaly score $s(\mathbf{x})$ for any $\mathbf{x}$:

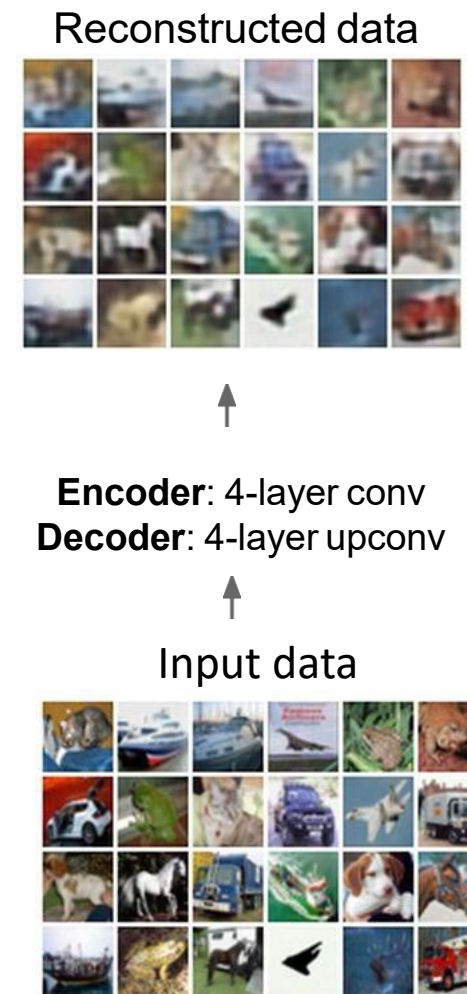
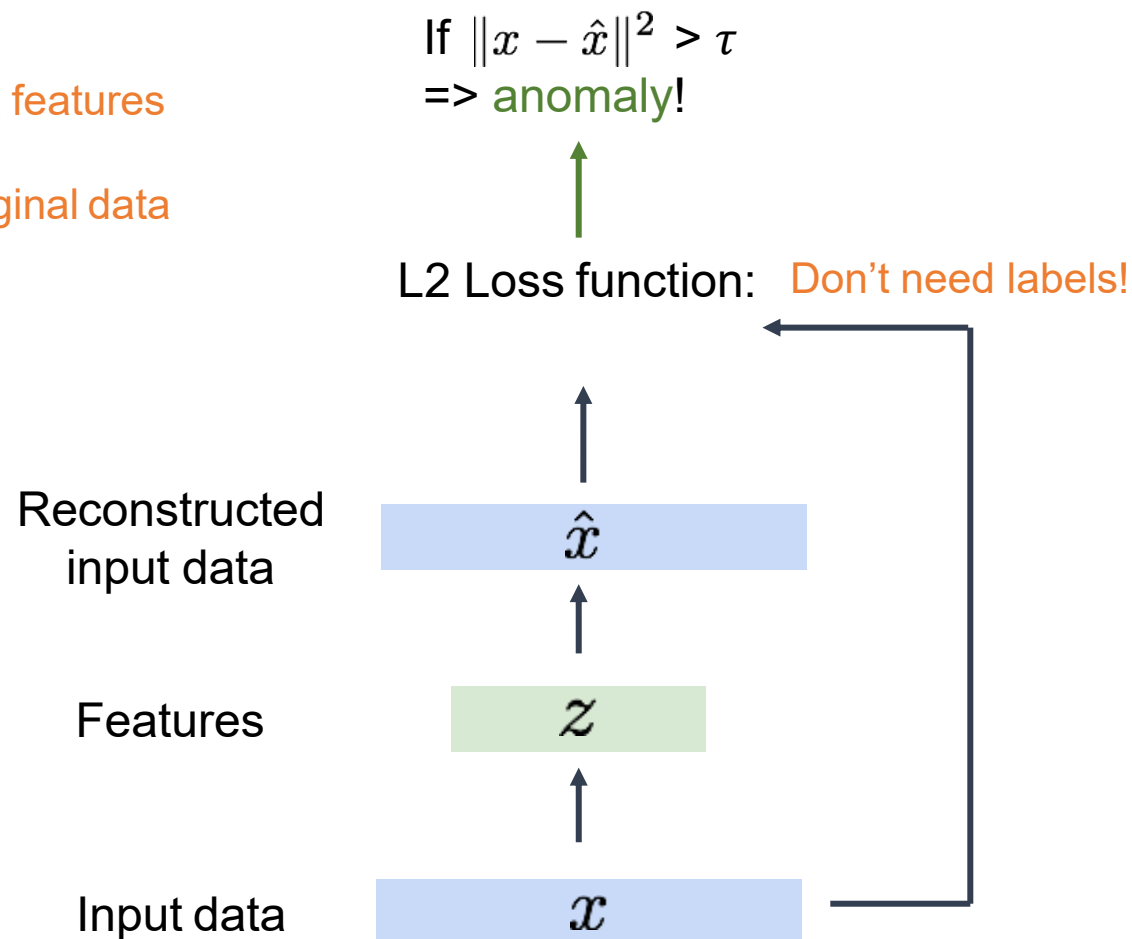
$$s(\mathbf{x}) = \|\hat{\mathbf{x}}_i - \mathbf{x}_i\|.$$



Encoder, decoder examples: PCA, Autoencoder

Autoencoder. General idea

Train such that features
can be used to
reconstruct original data



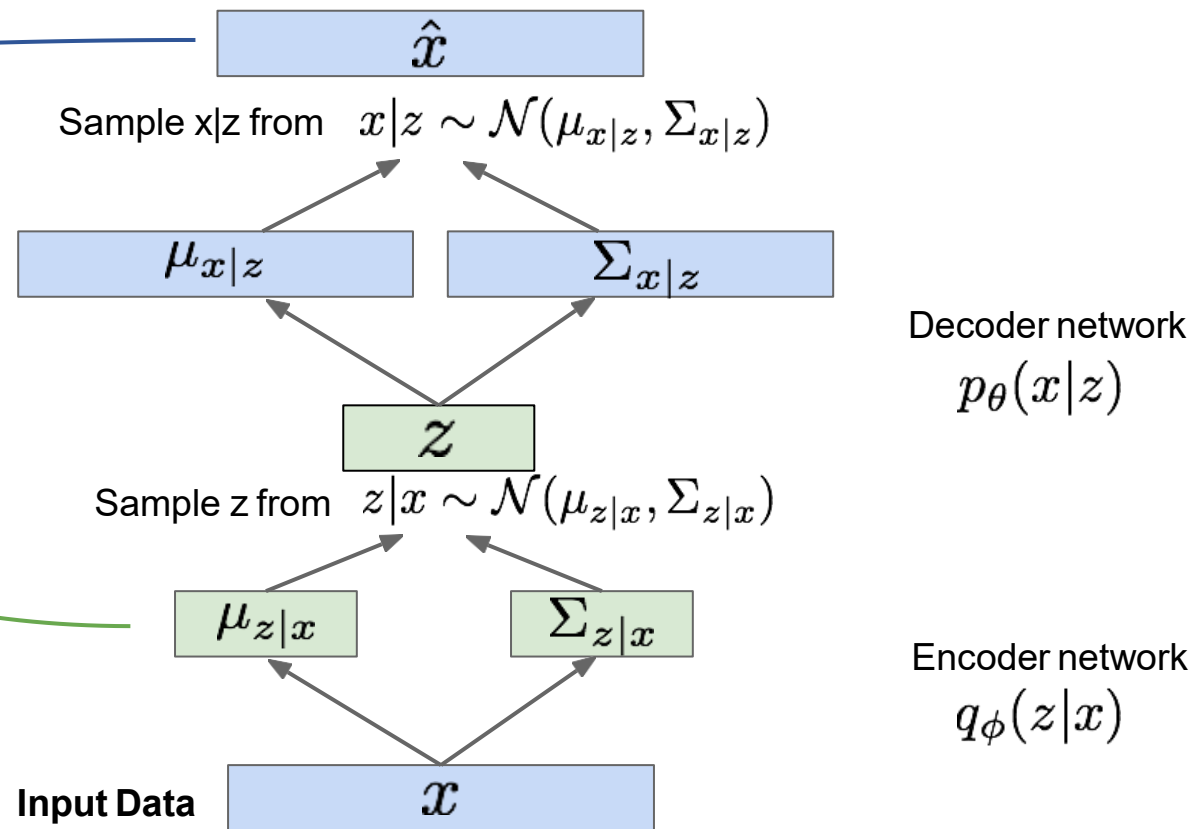
Variational Autoencoder

We maximize the likelihood lower bound

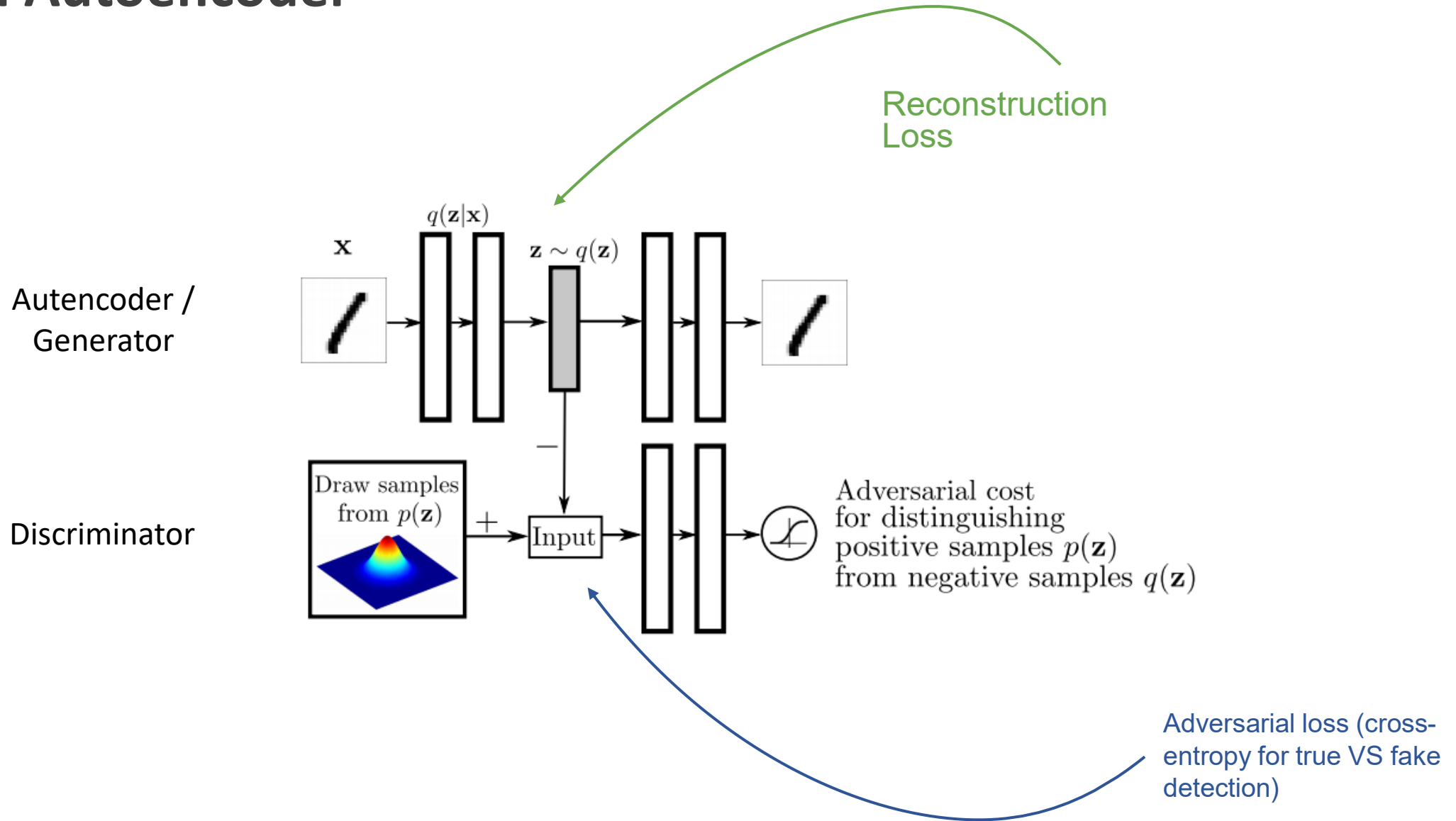
Maximize likelihood of original input being reconstructed

$$\underbrace{\mathbb{E}_z \left[\log p_\theta(x^{(i)} | z) \right] - D_{KL}(q_\phi(z | x^{(i)}) || p_\theta(z))}_{\mathcal{L}(x^{(i)}, \theta, \phi)}$$

Make approximate posterior distribution close to prior

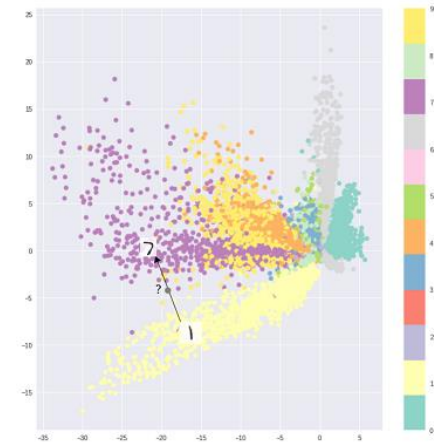


Adversarial Autoencoder

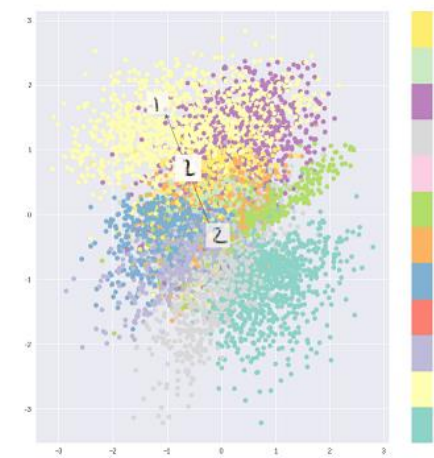


Why variational autoencoder?

- **Just autoencoder**
Sometimes we need just nonlinear PCA
The latent space may not be continuous
or allow easy interpolation.
- **Variational autoencoder**
If you want precise control over your latent representations and what you would like them to represent, then choose VAE.
Sometimes, precise modeling can capture better representations



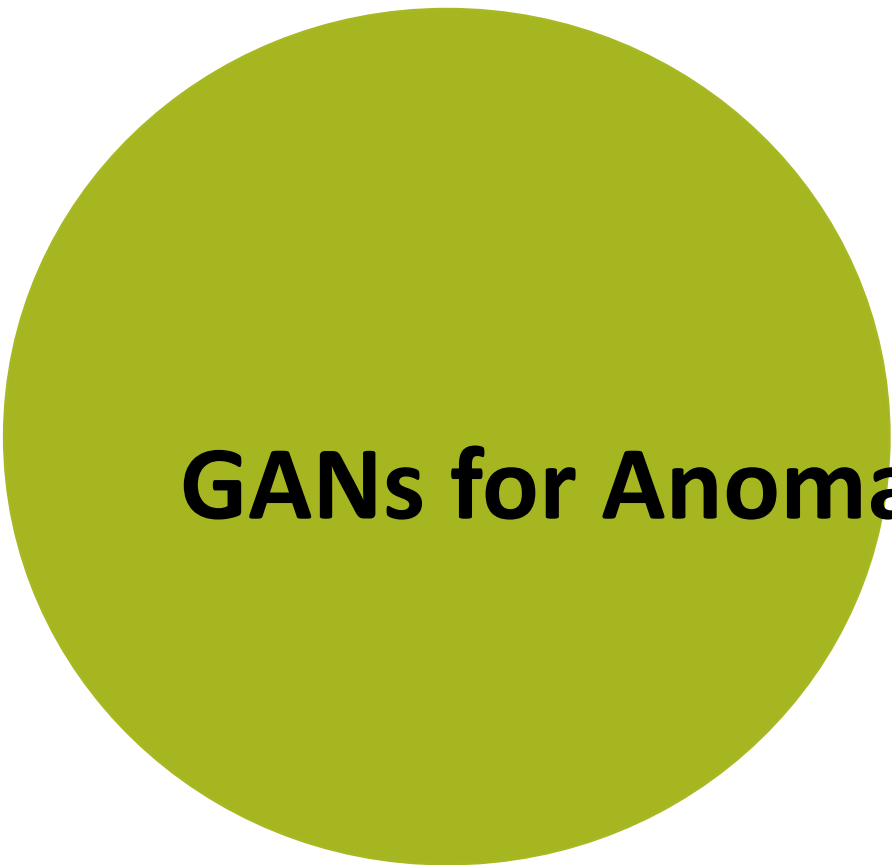
AE



VAE

<https://towardsdatascience.com/intuitively-understanding-variational-autoencoders-1bfe67eb5daf>

<https://arxiv.org/abs/1506.02216>



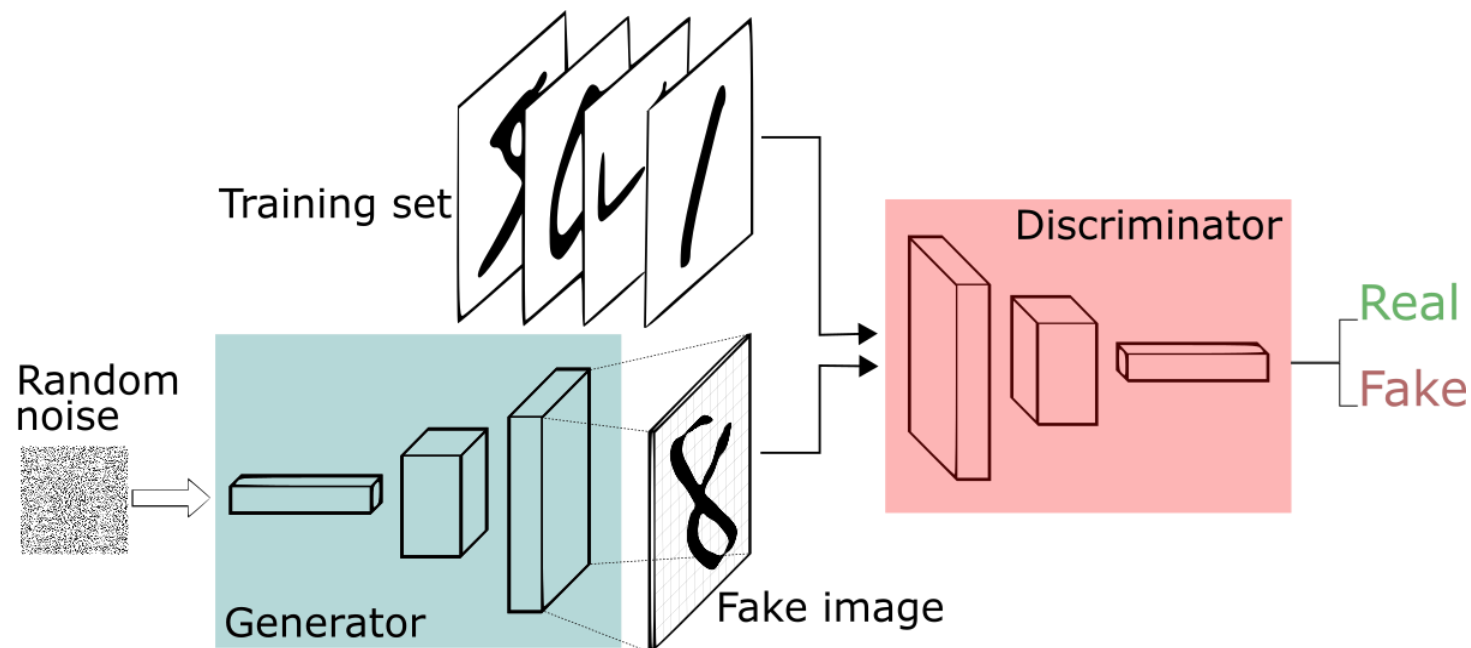
GANs for Anomaly Detection

Generative Adversarial Network

Generator network G: try to fool the discriminator by generating real-looking images

Discriminator network D: try to distinguish between real and fake images

Train jointly in **minimax game!**



Generative Adversarial Network: formulas

Minimax objective function:

$$\min_{\theta_g} \max_{\theta_d} \left[\mathbb{E}_{x \sim p_{data}} \log D_{\theta_d}(x) + \mathbb{E}_{z \sim p(z)} \log(1 - D_{\theta_d}(G_{\theta_g}(z))) \right]$$

Discriminator outputs likelihood in (0,1) of real image

Discriminator output for real data x Discriminator output for generated fake data G(z)

- *Discriminator* with parameters θ_d wants to **maximize objective** such that $D(x)$ is close to 1 (real) and $D(G(z))$ is close to 0 (fake)
- *Generator* with parameters θ_g wants to **minimize objective** such that $D(G(z))$ is close to 1: the discriminator is fooled into thinking generated $G(z)$ is real

GAN: example for novelty detection

The model is trained using images of penguins

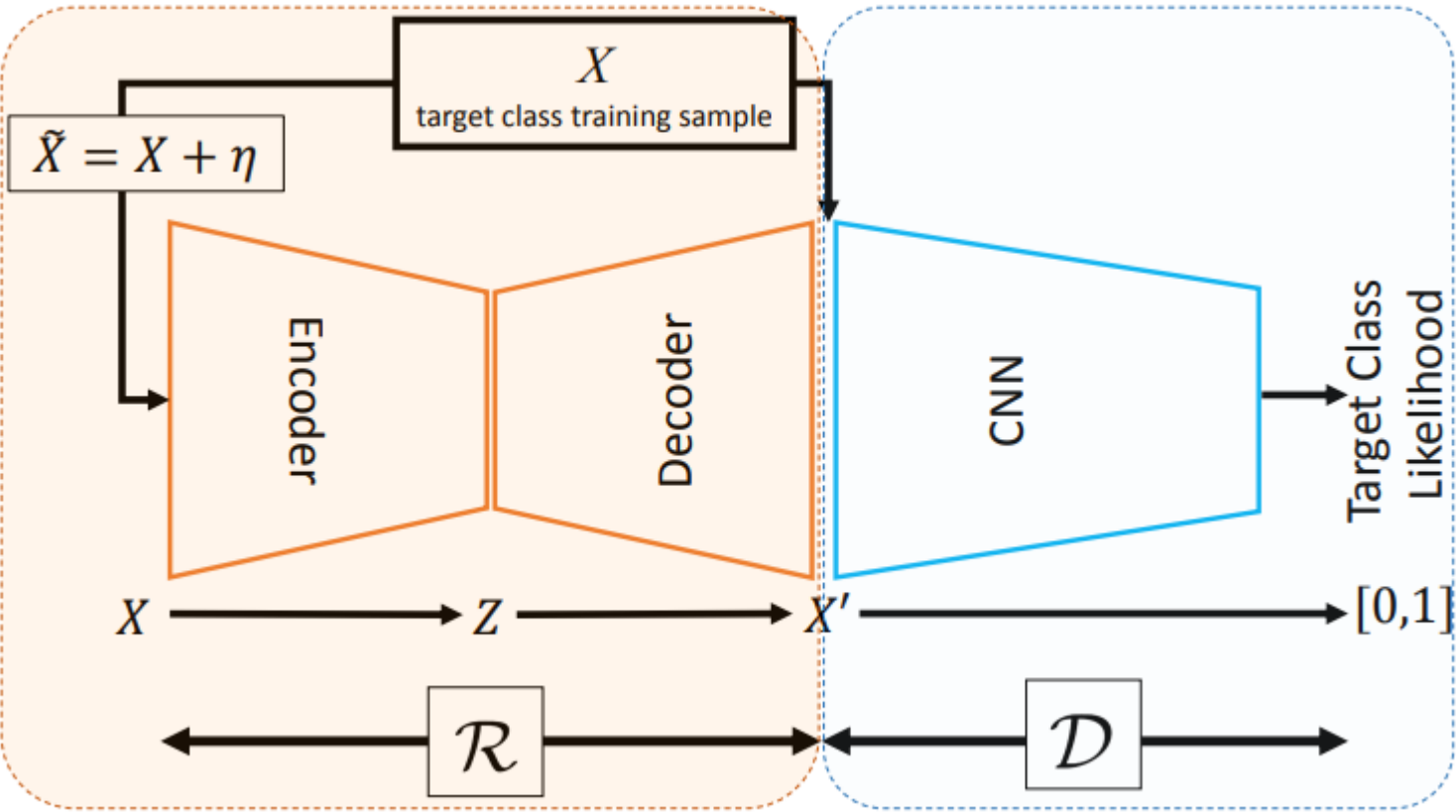


- If we use noisy inliers and pass them to “autoencoder” $\mathcal{R}$ NN, we get enhanced images as the output
- If we use outlier sample instead, the output of $\mathcal{R}$ is distorted

	Noisy Inlier Samples		Outlier Samples	
X				
$\mathcal{R}(X)$				
$\mathcal{D}(X)$	0.75	0.72	0.53	0.27
$\mathcal{D}(\mathcal{R}(X))$	0.85	0.91	0.25	0.10

GAN: example for novelty detection

Architecture

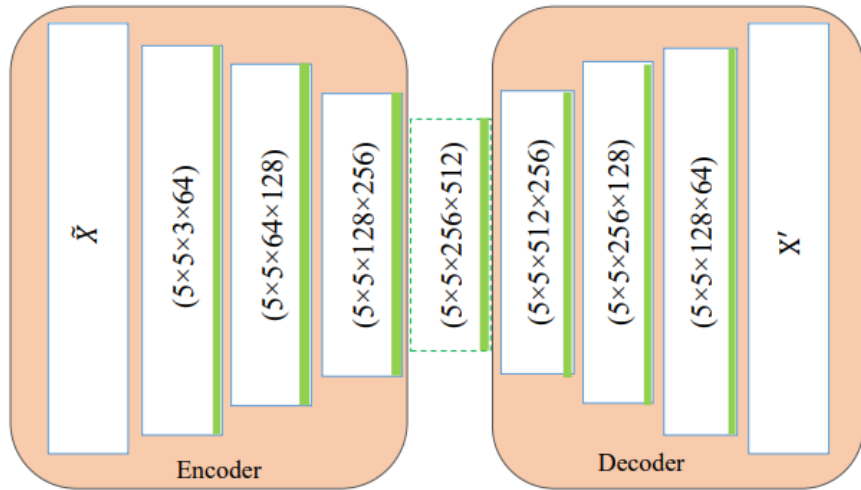


$$\mathcal{L} = \mathcal{L}_{\mathcal{R}+\mathcal{D}} + \lambda \mathcal{L}_{\mathcal{R}}$$

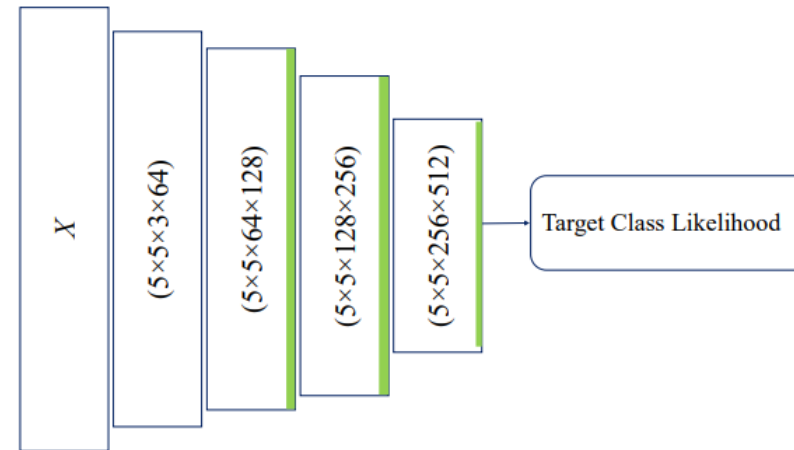
GAN loss

$$\mathcal{L}_{\mathcal{R}} = \|X - X'\|^2$$

Internal architectures



Autoencoder generator



Discriminator gives
anomaly score

<https://arxiv.org/abs/1802.09088>

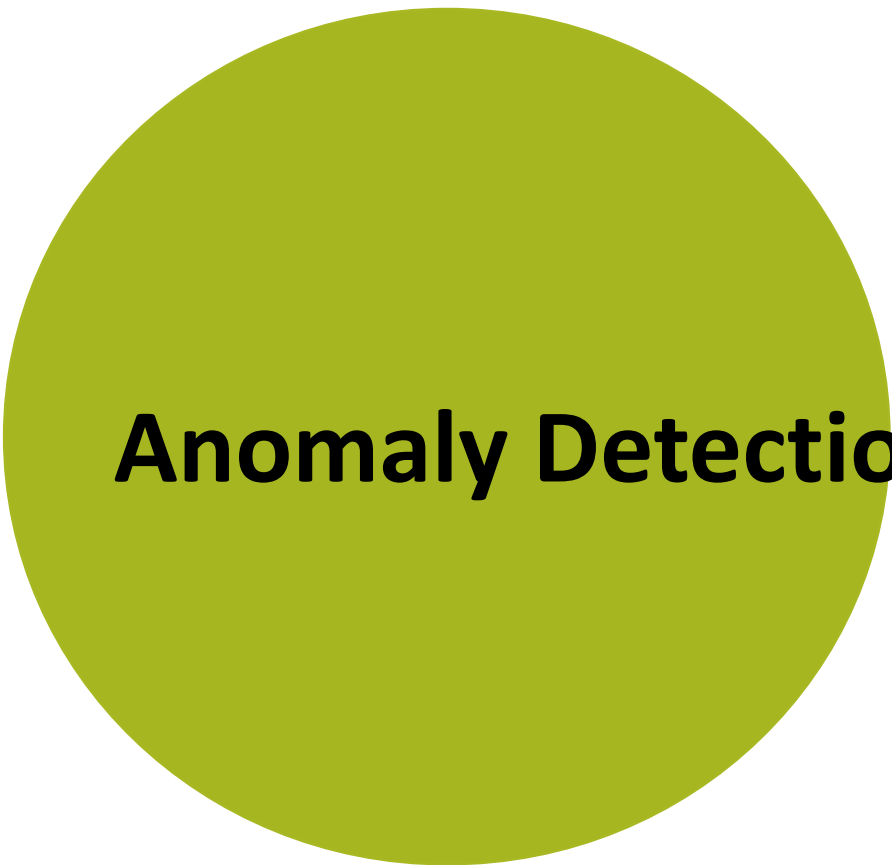
GAN: example for novelty detection

Anomaly score:

$$\text{OCC}_1(X) = \begin{cases} \text{Target Class} & \text{if } \mathcal{D}(X) > \tau, \\ \text{Novelty (Outlier)} & \text{otherwise,} \end{cases} \quad \text{PCA, VAE, AAE}$$

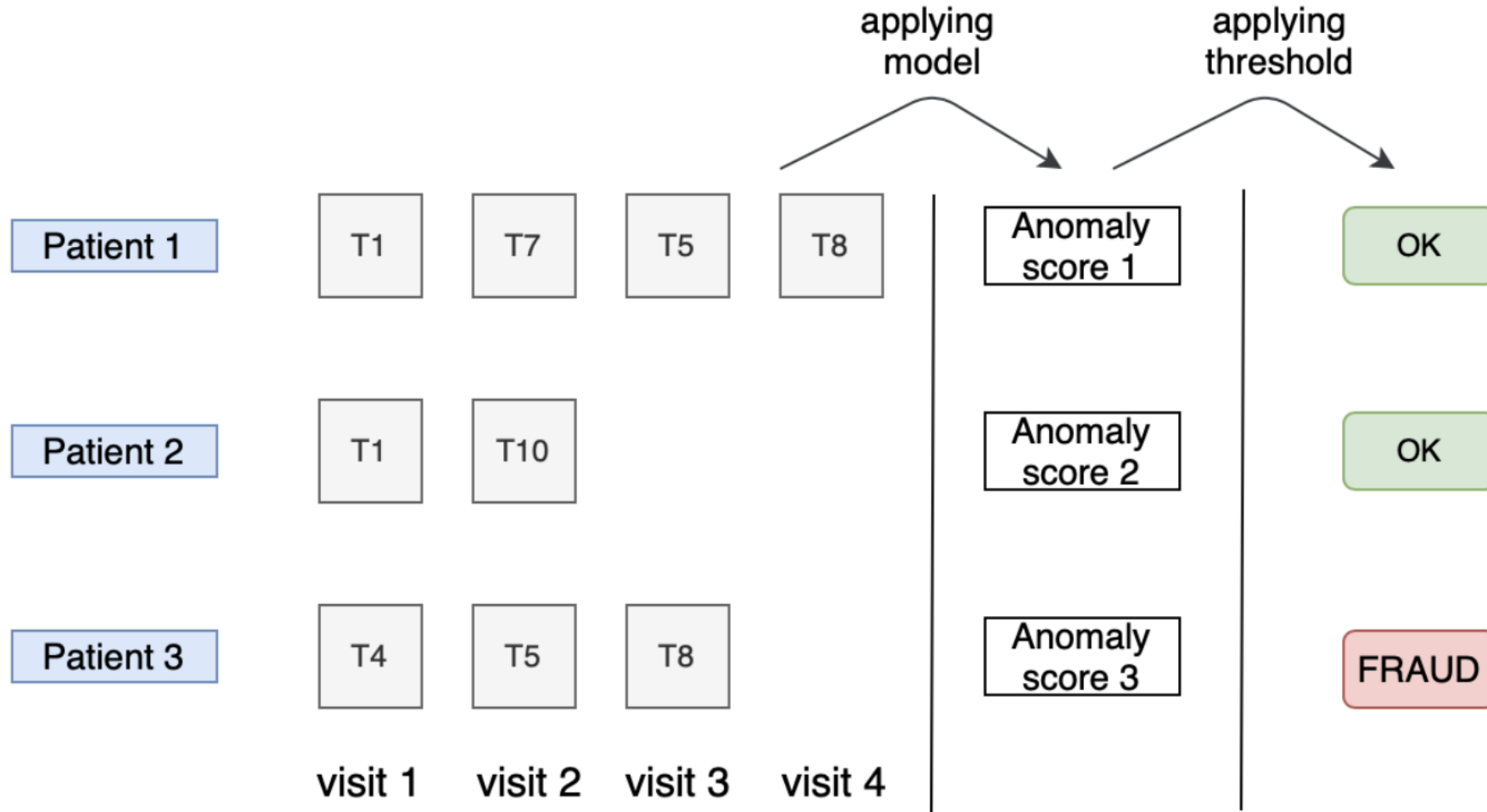
Anomaly score that utilizes encoder-decoder

$$\text{OCC}_2(X) = \begin{cases} \text{Target Class} & \text{if } \mathcal{D}(\mathcal{R}(X)) > \tau, \\ \text{Novelty (Outlier)} & \text{otherwise.} \end{cases}$$

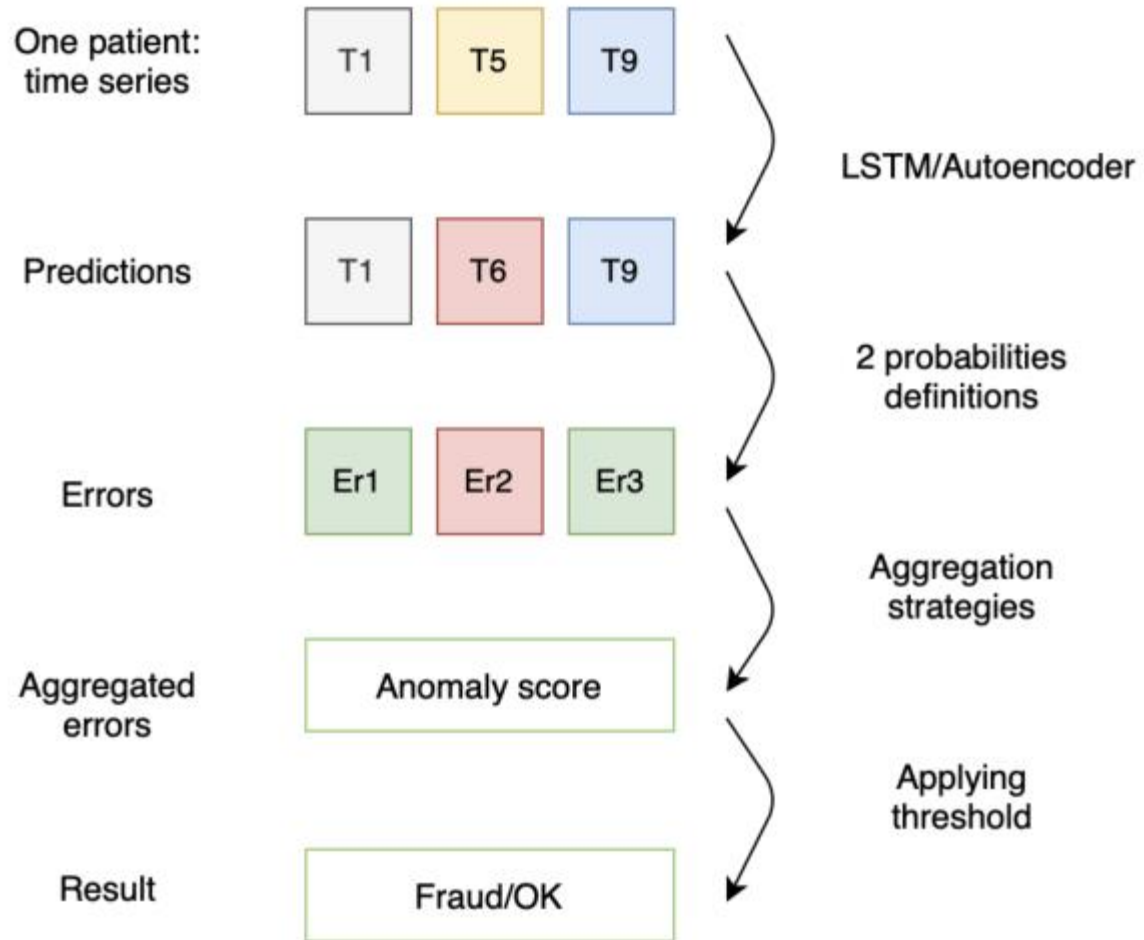


Anomaly Detection for Time Series

Anomaly detection for sequential data: healthcare insurance



Anomaly detection for sequential data: healthcare insurance



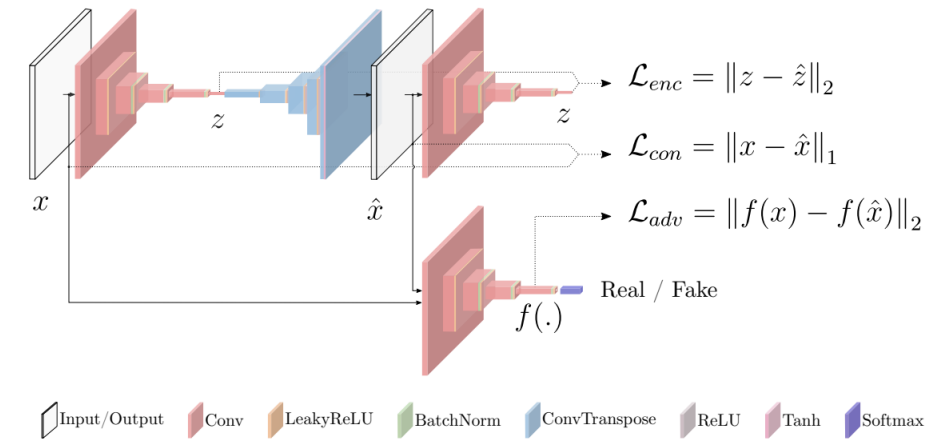
Anomaly and novelty for Time Series

Problems:

1. SotA techniques of anomaly detection are rarely used for classic time series
2. Available solutions don't take into account the statistical nature of Time Series

Proposed solution:

1. To develop a loss function for GAN-based anomaly detection for time series
2. To take into account requirements of statistical change point detection models: low number of false alarms and small detection delay
3. To develop new resampling techniques by learning data distribution



Li D. et al. MAD-GAN: Multivariate anomaly detection for time series data with generative adversarial networks // arXiv:1901.04997. – 2019.

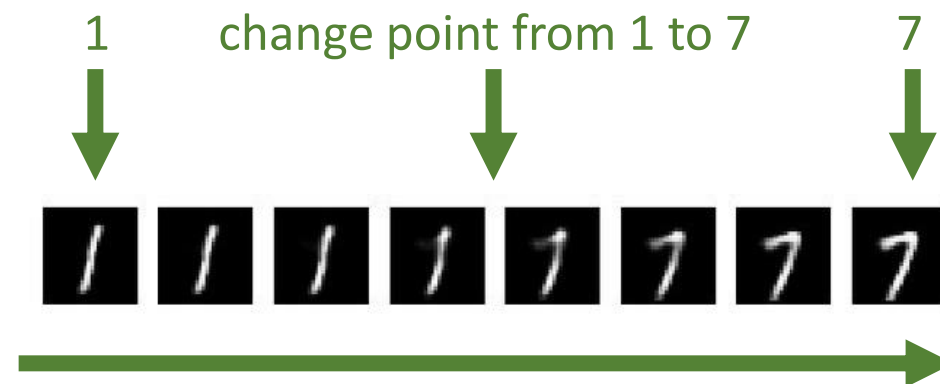
Change detection in semi-structured data

The change-point detection (CPD) model signals about time of change in the data distribution

Semi-structured data – sequences of semi-structured data (images, texts)

Goal: minimize Detection Delay & minimize number of False Alarms

Problems: Can't apply classic method for semi-structured data

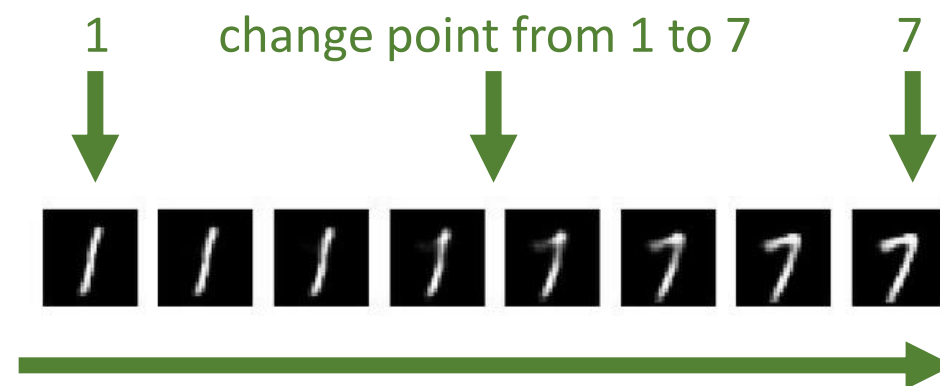


A sequence of MNIST images is an example of semi-structured data

Change detection in semi-structured data

Proposed solution:

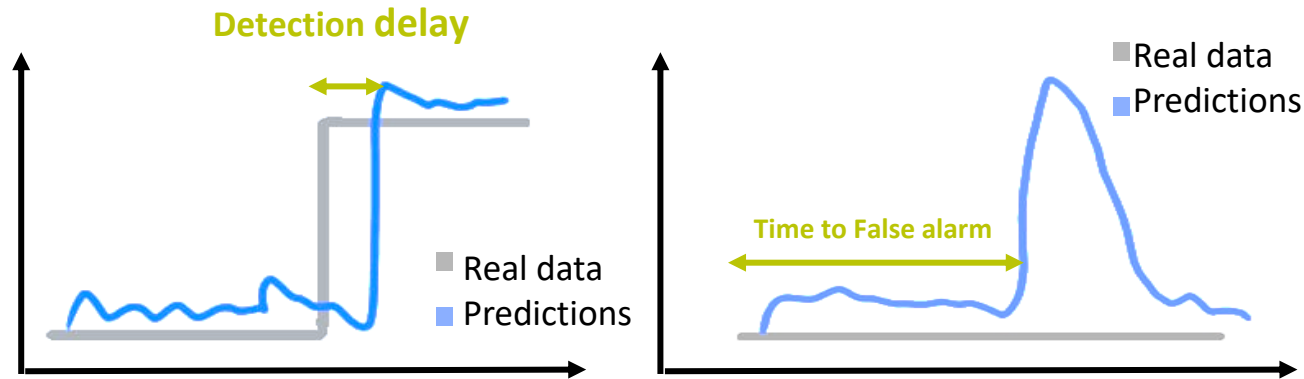
1. Develop a data embedding procedure
2. End2end methods based on statistical tests or detection outliers in embedded space – unsupervised anomaly detection
3. Develop new loss function for direct minimization of the problem specific metrics



A sequence of MNIST images is an example of semi-structured data

Our end2end approach

We concentrate on typical quality metrics for change-point detection: delay detection and mean time to False alarm.



We optimize lower bounds for these metrics:

- p_k is the model's change point probability at moment k ,
- T – hyperparameter that restricts the length of the considered sequence.

$$Loss_{detection_delay} = \sum_{t=\theta}^T (t - \theta) p_t \prod_{k=\theta}^{t-1} (1 - p_k) + (T + 1) \prod_{k=\theta}^T (1 - p_k),$$

$$Loss_{FP_delay} = 1 - \sum_{t=0}^{\theta} (t - \theta) p_t \prod_{k=0}^{\theta} (1 - p_k)$$

References

Change point detection (CPD). Basic knowledge and main statistical approaches:

- Shiryaev A. N. Stochastic disorder problems. – Springer International Publishing, 2019.
- Romanenkova E. et al. Real-Time Data-Driven Detection of the Rock-Type Alteration During a Directional Drilling //IEEE Geoscience and Remote Sensing Letters. – 2019.

Supervised CPD:

- Malhotra P. et al. Long short term memory networks for anomaly detection in time series //ESANN proceedings, 2015.
- Hundman K. et al. Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding //Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. – 2018

Conclusions

Factors to consider when choosing an Anomaly Detector

- Understand the nature of anomalies and the algorithm
- Few parameters
 - parameter-free the best
 - easy to tune; not too sensitive to parameter setting
- Fast runtime: can scale up to large datasets and high dimensional datasets
- Low space complexity
- Known behaviours under different data properties
- Can deal with different types of anomalies
- Its ability to deal with high-dimensional problems

https://federation.edu.au/_data/assets/pdf_file/0011/443666/ICDM2018-Tutorial-Final.pdf

Take-home messages

- Anomaly detection is a challenging problem
- Often problem-specific knowledge helps
- Common approaches are embedding-based and Isolation forest
- There are some time-series specific approaches: the problem is close to the change point detection problem

More references?

- Overview of anomaly detection for tabular data
<https://www.youtube.com/watch?v=12Xq9OLdQwQ>
- A collection of *awesome* anomaly detection papers
<https://awesomeopensource.com/project/hoya012/awesome-anomaly-detection>
- A collection of *awesome* anomaly detection resources
<https://github.com/yzhao062/anomaly-detection-resources>
- Link-based list of anomaly detection methods
<https://github.com/zhuyiche/awesome-anomaly-detection>

Backslides

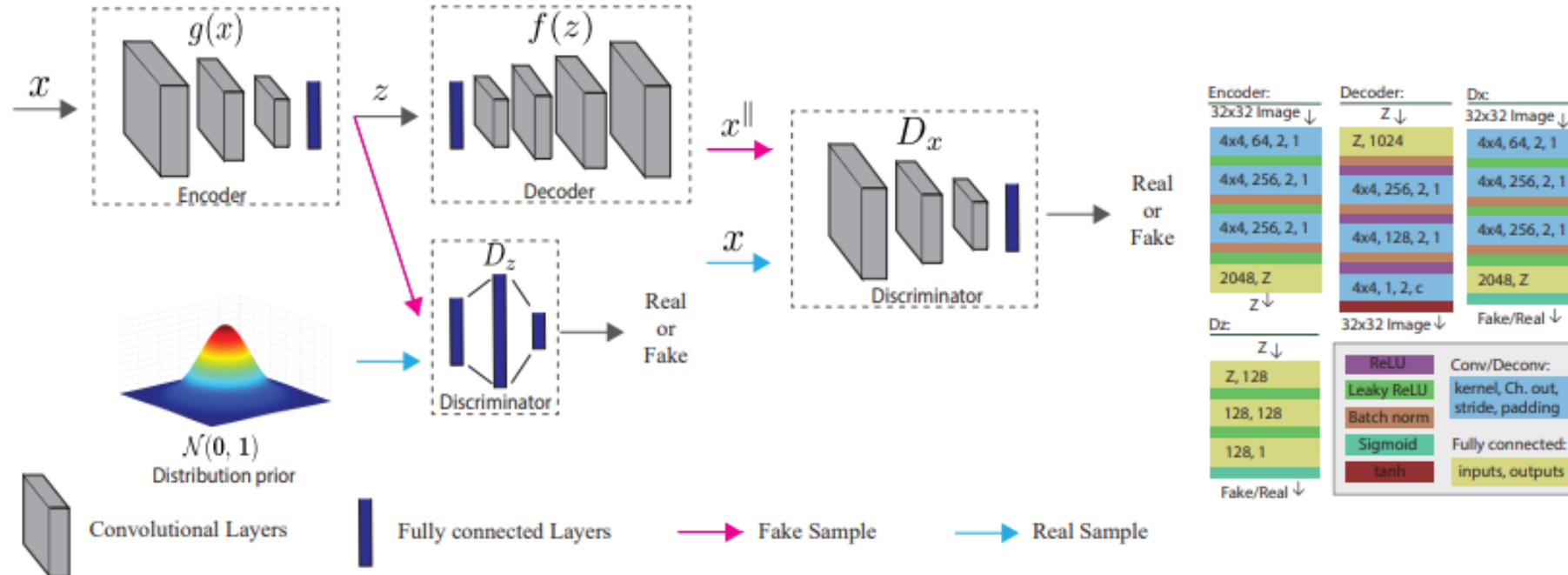
Model quality

	Normal Patches			Anomaly Patches	
X					
$\mathcal{R}(X)$					
$\mathcal{D}(X)$	0.15	0.19	0.32	0.35	0.44
$\mathcal{D}(\mathcal{R}(X))$	0.44	0.64	0.56	0.20	0.30

	CoP [32]	REAPER [22]	OutlierPursuit [50]	LRR [24]	DPCP [45]	R-graph [52]	Ours $\mathcal{D}(X)$	Ours $\mathcal{D}(\mathcal{R}(X))$
AUC	0.905	0.816	0.837	0.907	0.783	0.948	0.932	0.942
F_1	0.880	0.808	0.823	0.893	0.785	0.914	0.916	0.928
AUC	0.676	0.796	0.788	0.479	0.798	0.929	0.930	0.938
F_1	0.718	0.784	0.779	0.671	0.777	0.880	0.902	0.913
AUC	0.487	0.657	0.629	0.337	0.676	0.913	0.913	0.923
F_1	0.672	0.716	0.711	0.667	0.715	0.858	0.890	0.905

Adversarial autoencoders help

- Construct anomaly score $s(x)$ using data
- Signal about anomaly if anomaly score is greater than some threshold t



<https://papers.nips.cc/paper/7915-generative-probabilistic-novelty-detection-with-adversarial-autoencoders.pdf>

Results for InDID

Dataset: sequences with images from MNIST. There are sequences with (e.g. from 1 to 7) and without (e.g. from 1 to 1) change point.

We compare LSTM and fully connected neural network (FNN) architectures, as well as binary cross entropy loss (BCELoss) and our proposed loss.

LSTM with proposed loss function has a better Pareto frontier with respect to the mean detection delay and the mean false positive delay.

